Upper Tana-Nairobi Water Fund Monitoring and Evaluation Plan

By Craig Leisher, Justus Makau, Fred Kihara, Anthony Kariuki, John Sowles, David Courtemanch, George Njugi, and Colin Apse

Table of Contents

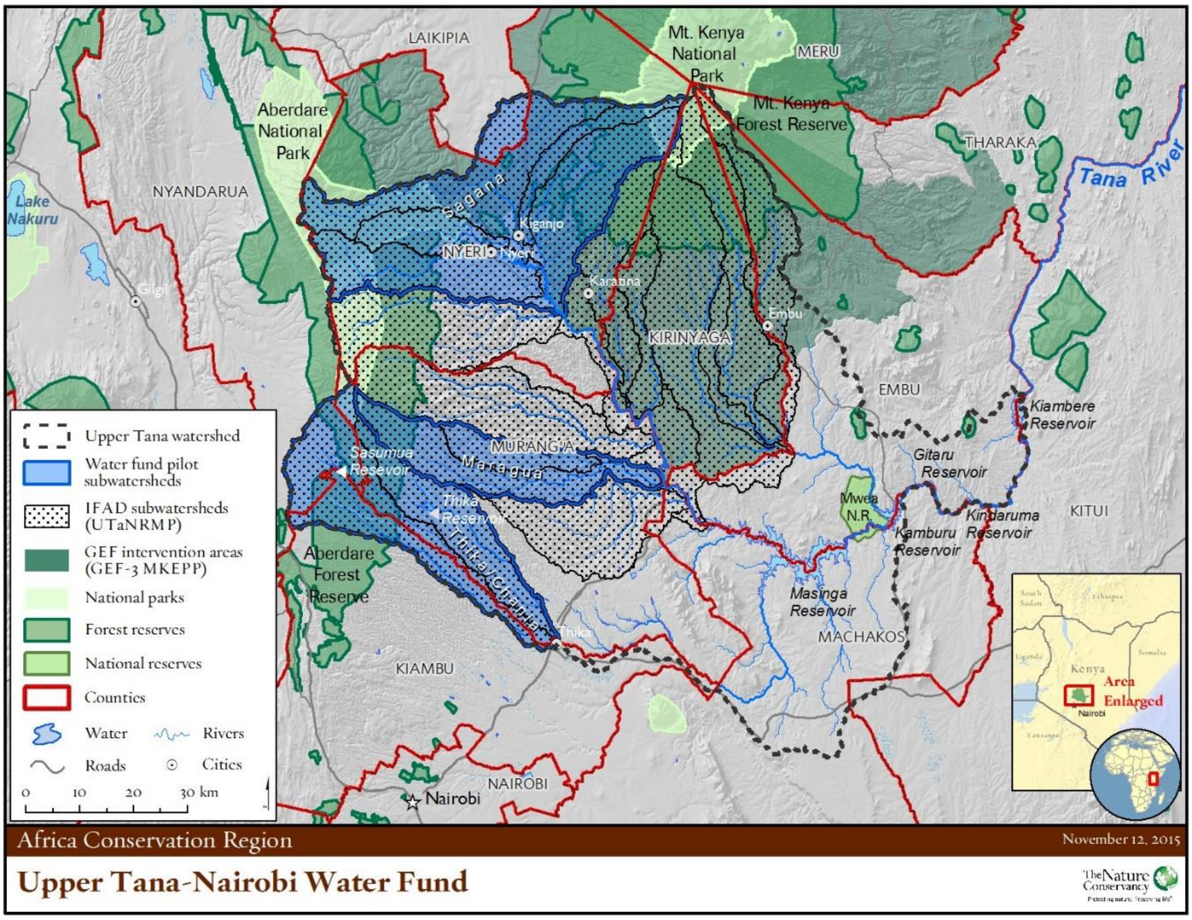
Acronyms

ARCT	Anne Ray Charitable Trust
BACI	Before-After Control-Impact
CBD	Conventional on Biological Diversity
DFID	Department for International Development
EX-ACT	Ex-ante carbon balance tool
GEF	Global Environmental Facility
GHG	Greenhouse gas
ICRAF	World Agroforestry Centre
IFAD	International Fund for Agricultural Development
IFPRI	International Food Policy Research Institute
INRM	Integrated Natural Resource management
KCEP-CRAL	Kenya Cereal Enhancement Programme - Climate Resilient Agricultural Livelihoods Window
LDSF	Land Degradation Surveillance Framework
M&E	Monitoring and Evaluation
MPAT	Multidimensional Poverty Assessment Tool
NCWSC	Nairobi City Water & Sewerage Company
NTU	Nephelometric Turbidity Units
PEMS	Program Evaluation and Monitoring System
PMU	Project Management Unit
RAPTA	Resilience, Adaptation Pathways and Transformation Assessment
RIMS	Results and Impact Management System
SLM	Sustainable Land management
TNC	The Nature Conservancy
UNCCD	United Nations Convention to Combat Desertification
UNFCCC	United Nations Framework Convention on Climate Change
UPS	United Parcel Service
UTaNRM	Upper Tana Catchment Natural Resources Management Project
UTNWF	Upper Tana-Nairobi Water Fund
WEAI	Women Empowerment in Agriculture Index
WRMA	Water Resources Management Authority

Pure water is the world's first and foremost medicine.
-- Slovakian proverb

Summary

Upper Tana-Nairobi Water Fund GEF Project



Project Summary Sheet

Country	Kenya			
Project Name	Upper Tana-Nairobi Water Fund			
Signing	**Effectiveness**	**Mid-Term Review**	**Project Completion**	**Project Closure**
6th October 2016	6th October 2016	2018-19 FY	30th June 2021	31st December 2021

Project Financing (USD '000)

Component	GEF	TNC	Private sector	NGOs & Counties	Beneficiaries	Total
Water Fund Management Platform Institutionalized	1,600	0	6,000	0	0	**7,600**
Improved upper tana catchment ecosystem supporting livelihoods & economic development	4,300	500	3,000	9,400	1,500	**18,700**
Robust KM and L systems implemented	1,000	300	1,000	2,400	0	**4,700**
Project Management	300	2,200		100		**2,600**
Total	**7,200**	**3,000**	**10,000**	**11,900**	**1,500**	**33,600**

Number of Beneficiaries

Total	Direct	Indirect	Women	Men
94,500	94,500	-	49,140	45,360

This document lays out the monitoring and evaluation plan for the GEF-funded Upper Tana-Nairobi Water Fund project and the Coca-Cola Africa Foundation grant. It will be updated annually.

Monitoring and evaluation are critical for the sustainability of a water fund. If the water fund fails to demonstrate in a systematic and rigorous way the benefits of watershed management, then the water fund may lose political, social, and financial support.

The project's theory of change can be summarised as: if we implement soil and water conservation activities in the upper Tana watershed, then people downstream at the 'tap' will have better water security, and people upstream at the 'top' will have improved livelihoods and well-being. Conservation benefits from the project activities include conserving aquatic biodiversity and restoring natural forest in the upper Tana.

The project's theory of change drives the selection of output and outcome indicators. Output indicators measure project implementation progress while outcome indicators are results that are necessary to achieve the intended impact of the water fund. The project has 21 output indicators and 13 outcome indicators covering ecological, social, and economic focal areas.

Data sources for the indicators come from several sources. Water quality data come from 26 new or upgraded water monitoring stations. Household-level indicators come from 1,000 household interviews. Greenhouse gas estimates come from EX-ACT ('Ex-ante carbon balance tool') informed by the Land Degradation Surveillance Framework surveys. Changes in water consumption before and after water conservation activities will be measured using a sample of smallholder treatment and control farms. Changes in water quality measurements before and after project activities will be measured in four control and seven treatment micro-watersheds.

Data for indicators that include acres, households, and trees will be collected and analysed quarterly. Data on water quality will be analysed every six months. Data for all the other indicators will be collected and analysed annually.

Learning activities include establishing two knowledge centres, compiling evidence-based lessons learned, school awareness programmes, peer-learning groups, and attendance at workshops and seminars to share water fund knowledge.

The project will have a mid-term review and a terminal evaluation. The overall evaluation question is: Did the project's soil and water conservation activities contribute to improvements in water quantity and quality as well as improvements in upper Tana smallholders' wellbeing?

A Project Management Unit will have the primary responsibility for the monitoring and evaluation (M&E) activities, and the project's fulltime M&E Officer will lead this work. The M&E Committee of the Water Fund Board of Management will provide oversight for the M&E activities. The committee will also conduct field visits to inform themselves on the progress and communicate their observations to the trustees and project steering committee members.

What we chose to measure becomes how we define success. Here success equals lasting improvements for both people and nature in Kenya's upper Tana River.

Introduction

This is the monitoring and evaluation plan for the Upper Tana-Nairobi Water Fund five-year project funded by the Global Environmental Facility that started on 6 October 2016 and ends on 31 December 2021. It includes monitoring and evaluation components specified under the grant from The Coca-Cola Africa Foundation grant.

This plan lays out the rationale, strategies, and costs for the project's monitoring and evaluations and links these to the project's learning activities.

Monitoring and evaluation are critical for the sustainability of a water fund. If the water fund does not demonstrate in a systematic and rigorous way the benefits of watershed management, then the water fund may lose political, social, and financial support.

This plan follows the *Program Evaluation and Monitoring System* (PEMS) for The Nature Conservancy's Africa Region, the monitoring and evaluation guidance in the International Fund for Agricultural Development's (IFAD's) water fund project documents, and elements of the Coca-Cola proposal. It also draws on The Nature Conservancy's *Primer for Monitoring Water Funds* and Conservation by Design 2.0 *Guidance Document*.

The project's stated *goal* is that the Upper Tana-Nairobi Water Fund public-private partnership increases investment flows for sustainable land management and integrated natural resource management in the upper Tana River catchment.

The project's *development objective* is a well-conserved upper Tana River basin with improved water quality and quantity for downstream users (public and private), regular flows of water throughout the year, enhanced ecosystem services, and improved human well-being and quality of life for upstream communities.

The upper Tana watershed provides 95% of Nairobi's water and 50% of the country's electricity. Water quality issues in the wet season and water quantity issues in the dry season are the two primary challenges in the watershed. High sediment loads during the wet season come from farmland with inadequate soil conservation measures, road building, landslides, and rock quarries. When sediment concentrations are high, the water flowing into Nairobi is reduced because the water treatment process requires more time. The high sediment loads also cause greater sedimentation in Masinga reservoir which is at the top of a cascade of five hydropower dams, and this reduces dry-season storage capacity and power generation. Climate change is increasing the unpredictability of the rains, and during drought years, water rationing for Nairobi's 3 million+ residents is now a reality. During the dry-season, water abstraction by farmers from local streams and rivers further reduces base flows. Addressing these water quality and water quantity issues is the core of the water fund.

Theory of Change

A theory of change states the logic of how the project's ultimate outcomes/impact will be achieved; it shows the causal pathways and the means to the ends. Here we use the UK's Department for International Development 'IF/THEN/BECAUSE' theory of change framing.[1] The 'if' statements define the inputs or activities. The 'then' statements detail the results of

[1] Corlazzoli V, White J (2013): Practical Approaches to Theories of Change in Conflict, Security and Justice Programmes. Part II: Using Theories of Change in Monitoring and Evaluation. DFID

those inputs and focus on what are termed 'intermediate results'. Intermediate results are those that are individually *necessary* and collectively *sufficient* to reach a desired impact.[2] The 'because' statements list the underlying assumptions.

Narrative theory of change

Strategy 1: Institutionalise water fund platform

IF we build transparent and accountable leadership for the water fund; IF we establish the water fund as a registered charitable trust; IF the water fund is integrated with national and local policies and strategies on watershed management; and IF the project outputs are utilized to identify and propose policies and strategies to facilitate watershed management:

THEN we will have a water fund that is a lasting institution in Kenya and provides a model for other catchments nationally and internationally.

Strategy 2: Improve the upper Tana catchment's ability to support livelihoods, food security, and economic development

IF we terrace steep and very steep farmlands; IF we reduce soil erosion from rural roads and quarries; IF we encourage grass strip barriers in sloping fields near rivers; IF we promote vegetative buffers along riverbanks and riverbank stabilization; IF we establish bio-gas units in households; IF we promote agro-forestry; and IF we encourage water conservation:

THEN water quality and quantity in the upper Tana will improve, there will be greater water supplies for Nairobi residents, at least 21,000 upper Tana households will have improved food security and resilience to climate change, at least 100,000 ha will be under sustainable land management, greenhouse gas emissions in the upper Tana catchment will decrease by 10% in 2021 compared to the 2013 baseline, and the river basin's aquatic and terrestrial biodiversity will be better conserved.

Strategy 3: Implement knowledge management and learning systems

IF we share the learning experiences of the water fund with at least two other natural 'water towers' in Kenya to assess the feasibility for water funds in these water towers; and IF we empower local and national government institutions to better monitor global environment benefits via water quality monitoring stations and Land Degradation Surveillance Framework surveys; and IF we establish water fund information centres at the National Museums of Kenya and the Ministry of Environment and Natural Resources:

THEN national and local government will have improved capacity to adaptively manage watersheds, Kenya will be better able to meet its reporting obligations under the three Rio Conventions (UNFCCC, CBD and UNCCD), and there will be greater political and social support for watershed management.

This is BECAUSE the water fund needs a strong institutional platform to become an autonomous and self-sustaining institution, BECAUSE households are more likely to continue soil and water conservation activities if they see financial or food security benefits from the activities, and BECAUSE improving national and local governments' ability to monitor global environmental benefits will allow them to enact better policies for sustainable land management.

---

[2] USAID (2013): Technical Note: Developing Results Frameworks.

Graphic theory of change

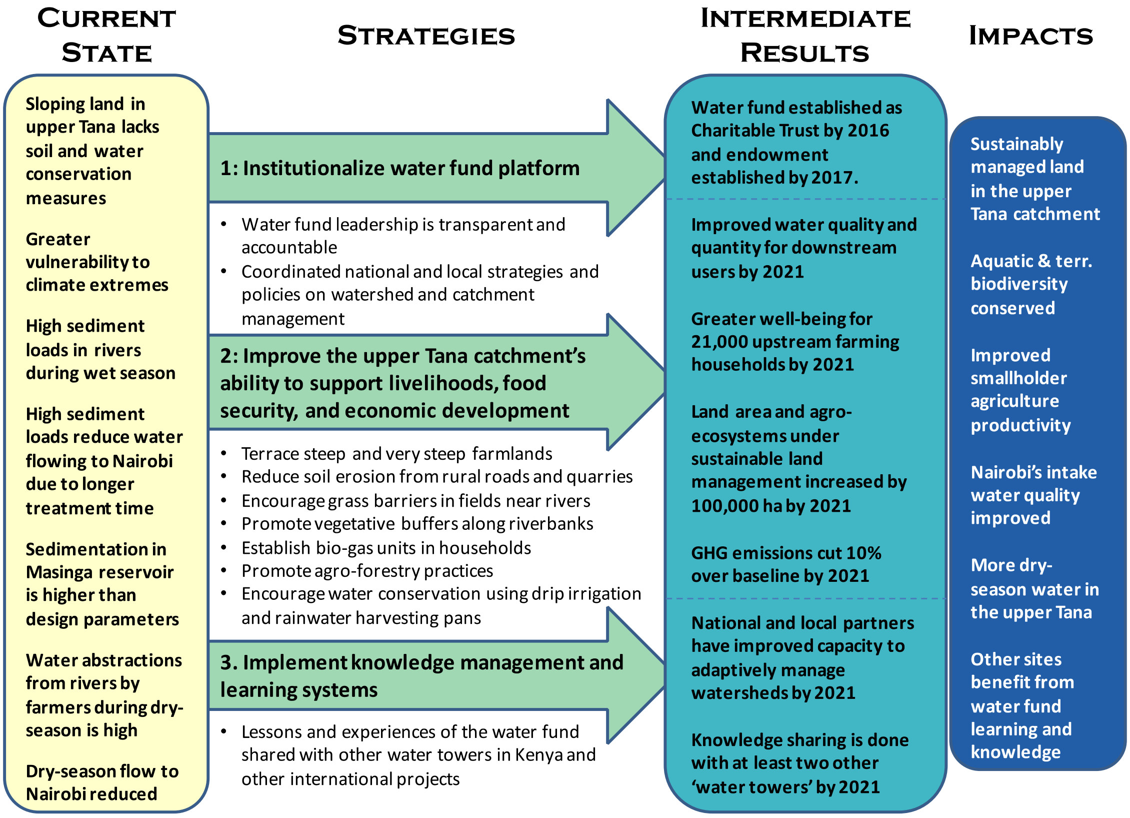


Figure 1: Theory of change graphic

Each intermediate result is tied to a project strategy or sub-strategy and is supported by output and outcome indicators. *Output* indicators are part of the monitoring, and *outcome* indicators are part of the evaluation.

Monitoring and evaluation are linked but do different things. Monitoring is ongoing and *describes* what is happening, while evaluation is periodic and *judges* how well it happened and what difference it made. Monitoring answers the question 'is the project doing things right' while evaluation answers the question 'is the project doing the right things'. Monitoring is first and foremost a management tool. Evaluations come in many varieties, but here we focus on a largely quantitative evaluation that measures if the project's strategies achieved the desired results.

Monitoring

Why we do monitoring

Monitoring gives project managers the information they need to adaptively manage the project. Adaptive management is about 'managing to learn' and 'learning to manage.' Adaptive management offers several benefits. This project is in a complex social-ecological system where when one social or ecological factor changes other factors change as well, so adaptive management takes advantage of the inherent uncertainties through learning. Adaptive management also reduces the need for extensive upfront planning because strategies can be revised as we learn.

Monitoring strategy

The plan's monitoring strategy is to collect data on output indicators under each of the three project strategies (a.k.a. project components). These data are the key input for the adaptive management of the project. Monitoring data will be aggregated electronically using a cloud-based monitoring and evaluation (M&E) system called DHIS2 (www.dhis2.org).

Monitoring results will be used to adaptively manage project activities. The donor-required Annual Monitoring Review will be the trigger for the adaptive management cycle of monitoring, reviewing, learning, and revising (Figure 2). There will be an annual 'pause and reflect' activity for the water fund.

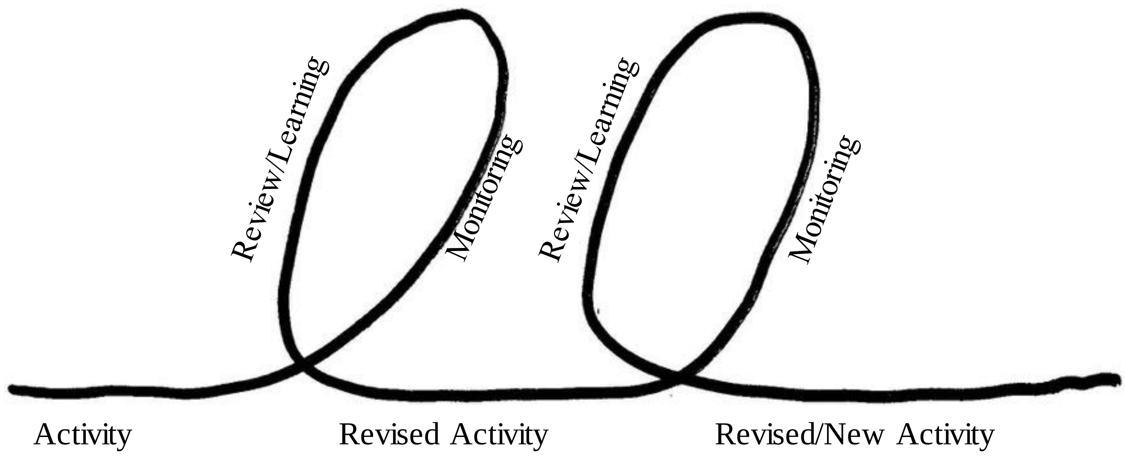


Figure 2: Adaptive management cycle

Output indicators

To track implementation progress for the project's three strategies, we will use the following output indicators (IFAD's 'first-level results'):[3]

Strategy 1

1. Status of the endowment fund (designed, legal documents drafted, submitted, and approved).
2. # of government policies and strategies that refer to the water fund as an incentive model.
3. Amount of funding in the endowment fund.
4. # of meetings held by the Project Steering Committee.
5. # of meetings held by the Board of Trustees.
6. # of national and county advisory committee meetings.

Strategy 2

7. # of people/individuals/household members receiving project services (IFAD's Results and Impact Management System—RIMS—1.8.2).[4]
8. # of hectares of land with Sustainable Land Management. This is a proxy for 'extent of land with rehabilitated or restored ecosystem services' (RIMS 1.1.17). This indicator comprises several sub-indicators:

[3] See Appendix 1 for detailed definitions of each indicator.
[4] Average household size in the project area is 4.9 based on household surveys in 2013 and 2014.

a. # of acres[5] with soil conservation measures
b. # of acres with water conservation measures
c. # of acres converted to perennial crops
d. # of acres converted to agro-forestry

9. # of people adopting technologies that reduce or sequester greenhouse gas emissions (RIMS 1.1.18). This will comprise:
    a. # of households (later converted to individuals) that add permanent grass barrier strips
    b. # of households that convert some or all of their land to a perennial crop
    c. # of household with new on-farm trees planted (agro-forestry)
    d. # of households installing bio-gas units

10. # of individuals involved in climate risk management, natural resources management, or disaster risk reduction activities (RIMS 1.6.10). This will comprise:
    a. # of households who implement soil conservation measures (disaggregated by type such as terracing or grass strips)
    b. # of households who implement water conservation measures
    c. # of households that convert some or all of their land to a perennial crop
    d. # of household with new on-farm trees planted

11. # of smallholder household members supported in coping with the effects of climate change (RIMS 1.8.6). Because all the soil and water conservation activities help smallholder households to cope with climate change effects, this indicator will count the same households as RIMS 1.8.2 (#7 above).
12. # of acres reforested.
13. # of households that installed a rainwater pan.
14. # of households that installed a drip irrigation system.

Strategy 3

15. # of Land Degradation Surveillance Framework surveys completed.
16. # of water monitoring stations installed or upgraded.
17. # of county development plans that reference the Multidimensional Poverty Assessment Tool survey results.
18. # of information sharing platforms established.
19. # of meetings, conferences, and seminars held at national, regional and international levels where water fund knowledge and learning were shared.
20. # of schools participating in school awareness programme.
21. # of new policies that give soil and water protection guidance on rural roads construction, quarries, afforestation, and riverine protection.

Data sources and methods

Data to track output indicators will come from the Project Management Unit for strategies 1 and 3 and from project implementation partners for strategy 2. The project implementation

[5] Kenya uses acres for land size. This will be converted to hectares after data collection.

partner organizations will track progress using either an Excel data entry form or an electronic data entry form on a smartphone or tablet linked to the project monitoring system via the internet. The data entry form will include the name of the person entering the data and the first names of those who participated in the project activity as well as their gender, age, and mobile phone number (for verification purposes), and the GPS location (latitude and longitude) of the activity.

Data collection frequency

Data for the indicators that include acres, households, and trees will be collected and analysed quarterly. Water quality data will be analysed every six months. Data for all the other output indicators will be collected and analysed annually.

Data quality assessment

A data quality assessment will assess the accuracy, validity, reliability, integrity and timeliness of the data as detailed below. Data quality assessments will be done annually for people-related data and semi-annually for water quality data and will comprise a review of implementation partner organizations' records, randomly selected spot checks to verify the reported data, and where relevant, phone calls to 10% of the participants to verify their data. When more than a small number of mistakes are found in the reported data, an audit of the partner's reported deliverables will be undertaken.

The project will also have an external financial audit every year, and this will include implementation partner organizations.

Data quality assessments will involve:

- Assessing the *accuracy* of the collected data by checking the correctness of the data and ensuring that deviations in data can be explained as well as ensuring measurement error is kept to a minimum. Accuracy is most likely to be secured if data is captured as close to the point of activity as possible.
- Evaluating data *validity* to ascertain that the data measures the result or outcome it is intended to measure.
- Appraising the *reliability* of data by analysing whether data collected over time are comparable and if trends are meaningful and the data allow for measurements of progress over time. Data should reflect stable and consistent data collection processes across collection points and over time.
- Assessing data *integrity* by checking that data quality assessments are integrated into data collection processes and procedures to ensure data are not erroneously reported or intentionally altered.
- Gauge *timeliness* by checking that data is captured as quickly as possible after the event or activity and is available for the intended use within a reasonable time period. Data should be available frequently enough to support management decisions.

Reporting requirements and dissemination plans

The required reporting outputs for project monitoring are:

- Six-month Progress Reports (as per Table 2 in the appendices of the project document) presenting the main achievements, issues and constraints of the reporting period, and information on financial and physical achievements in comparison with targets set in the Annual Work Plan and Budget as well as possible impact and outreach

- Annual Project Report (this will also be the Annual Monitoring Review, the principle instrument for reporting to the GEF Secretariat)
- Results and Impact Management System (RIMS) report
- Annual GEF Project Implementation Review
- The M&E plan for the year to be integrated into the Annual Work Plan and Budget

Reporting to IFAD using RIMS Excel template will be done on 31 March of each year except for the first year. The first RIMS report will be submitted by 31 October 2017 at the end of the first calendar year of effectiveness as per the project document. The second RIMS report will be submitted on 31 March 2018 and at the same time every year thereafter.

The Annual Project Report will be the primary document for reporting on project progress. It will be shared with implementation partner organizations and other including the Upper Tana Catchment Natural Resources Management Project (UTaNRMP), Kenya Cereal Enhancement Programme - Climate Resilient Agricultural Livelihoods Window (KCEP-CRAL), Anne Ray Foundation (ARF), United Parcel Service (UPS), and The Nature Conservancy (TNC) Africa's marketing and philanthropy staff.

Evaluation

Why we do evaluations

We do evaluations to assess project outcomes and answer the question: were the expected benefits realized? We also do project evaluations to learn what worked and what did not. Evaluations are where the longer-term learning of the project happens. The project's evaluation will focus on establishing the relevance, impact, sustainability, effectiveness and efficiency of the project.

Evaluation strategy

The project will use a performance evaluation strategy that measures before and after changes. It is challenging in a water fund to do a quasi-experimental impact evaluation because finding a counterfactual at the water-fund scale to show what would have happened without the project is problematic. The evaluation will be primarily quantitative but use qualitative tools to crosscheck the quantitative data and better understand the 'how' and 'way' of the quantitative data.

The overall evaluation question is: Did the project's soil and water conservation activities contribute to improvements in water quantity and quality as well as improvements in upper Tana smallholders' wellbeing?

The water *quantity* measurement strategy in the upper Tana watershed is not straight forward. The inter-annum variations in precipitation in the upper Tana[6] are almost certainly larger than the anticipated water savings from project activities. Moreover, climate change effects are likely to increase year-to-year variations in rainfall. Thus, the five years of the project is too short to see longer-term trends, and we are unlikely to measure sub-watershed changes in water

[6] Average annual rainfall in the upper Tana from 1996 to 2012 was 1,903 mm with a standard deviation of 463 mm.

quantity linked to the water fund activities. What we can and will do is measure water savings from project activities on smallholder farms. Measuring a sampling of farmers' baseline water abstraction from a river before the installation of a drip irrigation system or rainwater pans and afterwards is the approach we will use to show water quantity improvements from the project. In the sub-watershed where the project has 15 water monitoring stations, the water quantity focus will be on low-flow measurements.

The water *quality* strategy focuses on sub-watersheds and micro-watersheds. The sub-watersheds provide information on water resource conditions at a regional or landscape level and are useful to track system-wide changes in sediment loss/transport, water supply and use, and ecological flows over time. The micro-watersheds (<10 km² in area) test the efficacy of specific soil erosion control practices at the farm level and is similar to an 'edge of field' monitoring approach. Control and treatment micro-watersheds were matched by soil type, microclimate, no major construction areas, and distance from roads. The strategy is to establish the relative behaviour between control and treatment sites for at least one year prior to treatment and then see how that relationship changes after implementation of project activities. The project is supporting 11 micro-watershed water monitoring stations.

The challenges of a BACI design for a water fund's water quality monitoring

A Before-After, Control-Impact (BACI) design is a robust way to measure the impact of a project. Yet finding control sites that are similar ecologically, socially and economically to the water fund's impact/treatment sites is challenging. The Tana is a large river with a large catchment, and there is no comparable river in Kenya, so a catchment-level control is not possible. The Tana's sub-watersheds are also large, and the known variations in rainfall, crops grown, land slope, and land cover preclude controls at the sub-watershed level. Also the larger the watershed, the longer the time lag to detect water quality changes. It is at the micro-watershed level that control sites are possible. Micro-watersheds are less than 10 km² in area, and the stream can be waded except in high-flow conditions. Using matched micro-watersheds for treatment and control sites is the strategy adopted by this water fund. Alternatives to control sites are to measure water indicators upstream and downstream of treatment sites to assess net spatial change over time in water quality and quantity or to use an ecologically intact micro-watershed as a reference site and compare its water quality with the treatment sites. See TNC's *Primer for Water Fund Monitoring* for more on these alternatives.

Outcome indicators

To track the outcomes of the project, we will use the following indicators (IFAD's 'second-level results'): [7]

1. Effectiveness: increased ability of people to manage environmental and climate-related risks (RIMS 2.6.5).
2. Tonnes of greenhouse gas emissions (CO_2) avoided and/or sequestered (RIMS 2.1.9).
3. Project lessons reflected in government policies, strategies or programmes.
4. # of days per year with maximum turbidity measuring less than 5, more than 200 (threshold for switching from alum to more expensive polymers to clarify water), and more than 2,000 NTUs[8] at Nairobi's water intake.
5. Turbidity (NTU) in dry and wet seasons for both sub-watersheds and micro-watersheds.

[7] The indicators are defined in detail in Appendix 1 of this plan.
[8] NTU = Nephelometric Turbidity Units.

6. Sediment load (kg/year) in dry and wet seasons for water flowing into Masinga reservoir from the Maragua and Sagana Rivers.
7. % change in water abstracted from a river by smallholder farming households before and after installing drip irrigation and/or a rainwater pan.
8. % change in crop productivity before and after installing drip irrigation and/or a rainwater pan.
9. % of households with improved Multidimensional Poverty Assessment Tool score.
10. % of households saying permanent vegetation cover on their farm has increased.
11. % of households saying soil erosion occurs on their land.
12. % change in average number of trees per acre on survey households' land.
13. # of river kilometres protected.

Data sources and methods

All the data for the above household-level indicators will come from the Multidimensional Poverty Assessment Tool (MPAT) surveys with water fund-specific questions added. For greenhouse gas estimates, the data will come from the EX-ACT ('Ex-ante carbon balance tool') informed by the Land Degradation Surveillance Framework (LDSF) surveys to be conducted by the World Agroforestry Centre (ICRAF). For the water quality and quantity indicators, the data will come from 26 monitoring stations supported by the project and operated by Nairobi City Water & Sewage Company and the Water Resources Management Authority (WRMA). The water quality data monitoring details are included in the appendices. A baseline Women Empowerment in Agriculture Index (WEAI) survey was done in 2015 with the assistance of IFPRI in two sub-watersheds of the upper Tana. If funding is available, this will be done again at the end line to measure changes over time.

In partnership with the Kenya Electricity Generating Company, we will collect turbidity data from water monitoring stations just upstream of the Masinga Reservoir on the Maragua and Sagana Rivers. The water fund does not work in all the rivers that flow into the Masinga Reservoir, so the monitoring focus is on the two rivers where the project is working.

Information about the policy activities of the project will be tracked by capturing of information about policy discussions and activities with stakeholders at the local and national level, in order to facilitate the construction of Stories of Change during the project's mid-term review and terminal evaluation.

The two RIMS second-level results (outcomes) will be collected at the baseline and at the end line. If the speed of project implementation warrants and funding allows, a mid-term MPAT will also be conducted.

Baselines

The socioeconomic baseline was completed in March 2017 and comprised the MPAT's 143 questions plus 30 more questions on various water fund and climate resilience indicators. All socioeconomic data are gender and age disaggregated.

The survey sample frame was the 11 micro-watersheds where we have water quality monitoring stations. By doing the socioeconomic baseline in the same areas where we are doing water monitoring, we will be better able to estimate the socioeconomic changes driven by soil and water conservation activities and how much these activities contributed to changes in water quality or quantity. It will also give the project the potential to build predictive models of the socioeconomic effects of various soil and water conservation activities. The micro-watersheds are clustered in four major sub-watersheds where the project is working so will provide reasonable geographic coverage. The trade-off is the socioeconomic data will not be representative of the county or the ward levels where we are working. We are trading breadth for depth. The project already has representative baseline randomized household surveys from all the major sub-watershed (n = 2,106), and measuring localized changes in treatment and control micro-watersheds allows the project to learn which activities have greater social and ecological benefits.

The socioeconomic baseline sample size was 1,000 households. This sample size was selected to give the study adequate power to detect smaller changes. With a sample of 1,000, an alpha of 5% and a power of 80%, a 5% change would be detectable for variables with a baseline prevalence rate of 9%.

For the baseline water *quality* measurements, we measured turbidity and total suspended solids in the 11 micro-watersheds (four control and seven treatment sites) for at least a full year prior to project activities in the micro-watersheds. This established the pre-project relationship between the control and treatment sites. We will also measure *quantity* at various water stages at the micro-watershed sites. However, because of seasonal and inter-annum flow variability, this data cannot be used to infer the effects of water conservation measures on flow.

For the baseline water *quantity* measurements, we will select 100 farmers who are scheduled to receive drip irrigation systems and 100 farmers scheduled to receive rainwater pans, and measure their water consumption and crop productivity for one year prior to installation and one year after. Because Kenya's rainfall varies greatly from year to year, a counterfactual is needed to measure what would have happened anyway. Thus, we will also simultaneously measure water consumption in 200 control households where we pay them a small amount to record their water use for two years. After two years, the controls will receive the drip irrigation systems and/or rainwater pans and continue to track their water use for another year.

Mid-term review

The mid-term review will be in April 2019 and will be led by IFAD. IFAD will prepare the Terms of Reference and hire the consultants, and the project will fund and support the review. It will examine progress to date and revisit the project strategies to assess if any underlying assumptions or conditions have changed. Specifically, the mid-term review will:

- Assess whether project activities are leading to the desired outcomes.
- Review the expenditures to date and reallocate resources as needed to optimize project outcomes.
- Examine differences between planned and actual results.
- Identify key accomplishments and challenges.
- Identify lessons learned thus far and make recommendations for any mid-term corrections that may be necessary for the project to best achieve its desired outcomes.

Terminal evaluation

The purpose of the terminal evaluation is to obtain an independent assessment of results achieved, lessons learned, and potential for replication. Specifically, the terminal evaluation will:

- Assess whether the project achieved the intended outcomes.
- Assess the likelihood that project benefits will be sustained.
- Document key successes, challenges, and lessons learned, and provide specific recommendations for similar projects.
- Assess the future sustainability of the project.

The socioeconomic final evaluation will start seven months before the project ends on 31 December 2021. It will use the same survey instrument with the same households (a 'panel' design) to decrease the risk of sampling bias and compare the baseline and end line levels for each indicator.

The socioeconomic final assessment will include a series of focus group discussions to explore the project outcomes in more detail. Focus group discussions will be held in project watersheds with farmers, female household heads, and youth. These discussion will specifically seek to identify possible alternative explanations for the observed changes to assess the likely causality of project activities.

The water quality and quantity assessment will summarize the changes in the micro-watersheds over time and compare control and treatment outcomes quantitatively. Both the socioeconomic and water assessments will be used to answer the project's evaluation question stated above.

A final evaluation mission led by IFAD will review the findings from the evaluations and assess the sustainability of the project outcomes. IFAD will prepare the Terms of Reference for the evaluation and hire the consultants, and the project will fund and support the evaluation.

Learning

The project has a specific learning agenda, and learning and adaptive management are two sides of the same coin. Strategy 3 includes a call to 'capture, document and disseminate UTNWF's lessons learned and foster policy dialogue on the interlinkages between implementation for multilateral environmental agreements and food security goals.' To accomplish this, the Project Management Unit will systematically identify and capture knowledge generated during project implementation and ensure that this knowledge is documented and shared. The knowledge will be captured and documented in the form of lessons learnt and good practices. It will be disseminated to stakeholders through appropriate communication channels to strengthen their capacities and improve performance.

The following tools and approaches will be used to promote learning within the project area and beyond:

- Evidence-based lessons learned on successful approaches for that could be adopted by others
- Documentaries (videos and pictures) on outcomes resulting from project activities
- Establishment of knowledge centres with the National Museums of Kenya and the Ministry of Environment and Natural Resources
- Posters, leaflets, and maps highlighting key conservation messages from the project
- School awareness programmes

The dissemination channels for the sake of sharing and learning will include:

- Seminars and workshops such as the GEF's Integrated Approach Programme on Fostering Sustainability and Resilience for Food Security in Sub-Saharan Africa
- Project website
- Social media
- Roundtable discussions
- Study/research reports
- National newspapers, TV and radio stations
- Peer-learning groups among key stakeholders. For example, farmer-to-farmer learning will be established to share knowledge and interact as a learning community on key areas of interest.
- Annual multi-stakeholder dialogue/reviews

Logical Framework

Hierarchy	Indicators				Means of Verification			Assumptions
	Description	**Baseline**	**Mid-term**	**End line**	**Source**	**Frequency**	**Responsibility**	
***Goal*: The Upper Tana-Nairobi Water Fund as a public-private partnership increases investments flows for sustainable land management and integrated natural resource management in the upper Tana catchment**	21,000 small holder farmer households with improved food-security, climate change adaptation and resilience capabilities (gender and age disaggregated) (RIMS 1.8.2)	Zero	30% over baseline	21,000 smallholder households	MPAT surveys	Baseline End line	PMU	National and county governments supportive of the WF concept
	Number of tonnes of greenhouse gas emissions (CO_2) avoided and/or sequestered (RIMS 2.1.9)	TBD	5 % reduction	10% reduction compared to baseline	LDSF surveys	Baseline End line	PMU/ ICRAF	A large shift to perennial crops among smallholders is viable
***Development objective*: a well-conserved upper Tana River basin with improved water quality and quantity for downstream users (public and private; maintaining regular flows of water throughout the year; enhancing ecosystem**	21,000 smallholder farmer households adopt climate-smart sustainable land management practices	Zero	30% over baseline	100%	LDSF surveys	Baseline End line	PMU/ ICRAF	Downstream water users (public and private) are interested in supporting upstream sustainable land management to improve water quality and quantity
	Improved base flow in rivers (M^3/sec)	TBD	10% above controls	30% above controls	Water monitoring stations	Baseline Mid-term review End line	PMU	
	Reduced sediment load in river basins in wet season (kg/l/sec)	TBD	5% reduction over controls	10% reduction	Water monitoring stations	Baseline Mid-term review End line	PMU	
Strategy 1: Water fund platform institutionalized								
***Outcome 1.1*: Multi-stakeholder and multi-scale platform supports policy development, institutional reform, and upscaling of integrated natural resource management (INRM) and sustainable land management (SLM)**	Water fund endowment is operational	Zero	Water fund by-laws legally established	Water fund capitalized	Legal documents, bank statements	Mid-term review Project end	PMU	Proposed Public Benefits Act supports water fund establishment
	Relevant government policies and strategies refer to the water fund as an incentive model	Zero	≥ 2 policies and strategies at national/ county levels	≥ 4 policies and strategies at national/ county levels	Official documentation records	Mid-term review Project end	PMU	Policies and strategies open for amendment and addition

Hierarchy	Indicators				Means of Verification			Assumptions
	Description	**Baseline**	**Mid-term**	**End line**	**Source**	**Frequency**	**Responsibility**	
Outcome 1.2: Policies and incentives support climate smart smallholder agriculture and food value chains in financially viable and sustainable watershed stewardships	Coordinated watershed management policies at county and national levels	Overlapping scales and responsibilities	Input into 3 CIDPs and sectoral strategies	3 CIDPs coordinated with the WRMA	WRMA records	Mid-term review Project end	PMU	Policy and strategy formulation at local, county and national level can be coordinated
	Smallholder people/ individuals/ household members receiving project services/incentives (RIMS 1.8.2)	Zero	30% of target	21,000 households	Water Fund implementation partners' records and disbursement records	Mid-term review Project end	PMU	Smallholder households are interested in joining incentive schemes
Strategy 2: Improve the upper Tana catchment's ability to support livelihoods, food security, and economic development								
Outcome 2.1: Increased land area, freshwater, and agro-ecosystems under INRM and SLM	Number of hectares of land brought under climate-resilient management	Zero	30,000 Ha	100,000 ha[9]	Project reports	Start up Mid-term review Project end	PMU	Smallholders are actively supporting INRM and SLM approaches
	Number of individuals provided with technologies that reduce or sequester greenhouse gas emissions (RIMS 1.1.18)	Zero	7,000 households	21,000 households	Project reports	Start up Mid-term review Project end	PMU	Land-use changes can be facilitated and are accepted
Strategy 3: Implement knowledge management and learning systems								
Outcome 3.1: Institutions capacitated to monitor Global Environmental Benefits	Global Environmental Benefits monitoring tools and protocols integrated with partner institutions	Zero	≥ 2 Land Degradation Surveillance Framework (LDSF) surveys updated/ completed. ≥ 10 water monitoring stations upgraded/ operational. WRMA prepared to house water-	LDSF surveys for at least 5 sub-watersheds updated/ completed. 26 monitoring stations upgraded/ operational. A water-quality database established and integrated into WRMA system.	LDSF reports; Project records	Mid-term review Project end	PMU	Institutional processes allow for integration of monitoring protocols

[9] As per Table 1 in the project document.

Hierarchy	Indicators				Means of Verification			Assumptions
	Description	**Baseline**	**Mid-term**	**End line**	**Source**	**Frequency**	**Responsibility**	
			quality database.					
Outcome 3.2: Monitoring and assessment framework supports the integration of climate resilience into policy making	Multidimensional Poverty Assessment Tool (MPAT) survey results referenced in county development plans	Zero	2 MPAT survey references	MPAT survey results referenced in ≥ 1 County Integrated Development Plan	County Integrated Development Plans, project reports, M&E records	Baseline Mid-term review End line	PMU	County development agencies open to new approaches
Outcome 3.3: Knowledge management and sharing of lessons learned is facilitated	Information sharing platfoms established	Zero	1 county-level info centre	1 county and 1 national-level info centre	Project reports	Mid-term review Project end	PMU	Partner organisations willing to establish and operate information centres
	Number of meetings, conferences, and seminars held at national, regional and international levels where water fund knowledge and learning were shared	Zero	Inputs/ presentations at ≥ 5 meetings at national, regional and international levels	Inputs/ presentations at ≥ 10 meetings at national, regional and international levels	Project reports	Mid-term review Project end	PMU	Opportunities for influencing dialogues present themselves
	Lessons learned out-scaled to at least 2 other catchment areas in Kenya	Zero	Lessons learned for scaling out are being complied	≥ 2 feasibility studies conducted	Project reports	Mid-term review Project end	PMU	Other water towers and relevant authorities interested and engage in feasibility studies

Implementation and Management of M&E Activities

Responsibilities

The Project Management Unit will have primary responsibility for the M&E activities, and the project's dedicated M&E Officer will lead this work. Much of the data will come from project implementation partners, so there will be a specific M&E person at each of the implementation partners who will collect and upload M&E data periodically. The M&E officer will also do the annual data quality assessments and will draft the monitoring reports. The Water Fund Manager will review the draft monitoring reports and edit them as needed before submitting them to the appropriate donor representative.

The M&E Committee of the Water Fund Board of Management will provide oversight for the M&E activities. It will also conduct field visits to inform themselves on the progress and communicate their observations to the trustees and project steering committee members.

Mechanism for updating the M&E Plan

Every year when the Annual Project Report is being prepared, the M&E team will review the existing M&E Plan and note areas where it needs to be updated. These recommended updates will be included in the Annual Project Report to be approved by IFAD.

Stakeholders

The Upper Tana-Nairobi Water Fund project includes a diverse group of stakeholders with an interest in the project's monitoring and evaluation as either data providers or data consumers (Figure 3). All of the data providers will be educated about their reporting obligations and a standardized form agreed for their reporting. This form can be electronic (the preferred option) or in Excel. The deadlines for data submission will be one month before the data are due to the donor.

Data consumers are organizations with an interest in project progress. These are the target audience for the monitoring and evaluation reports.

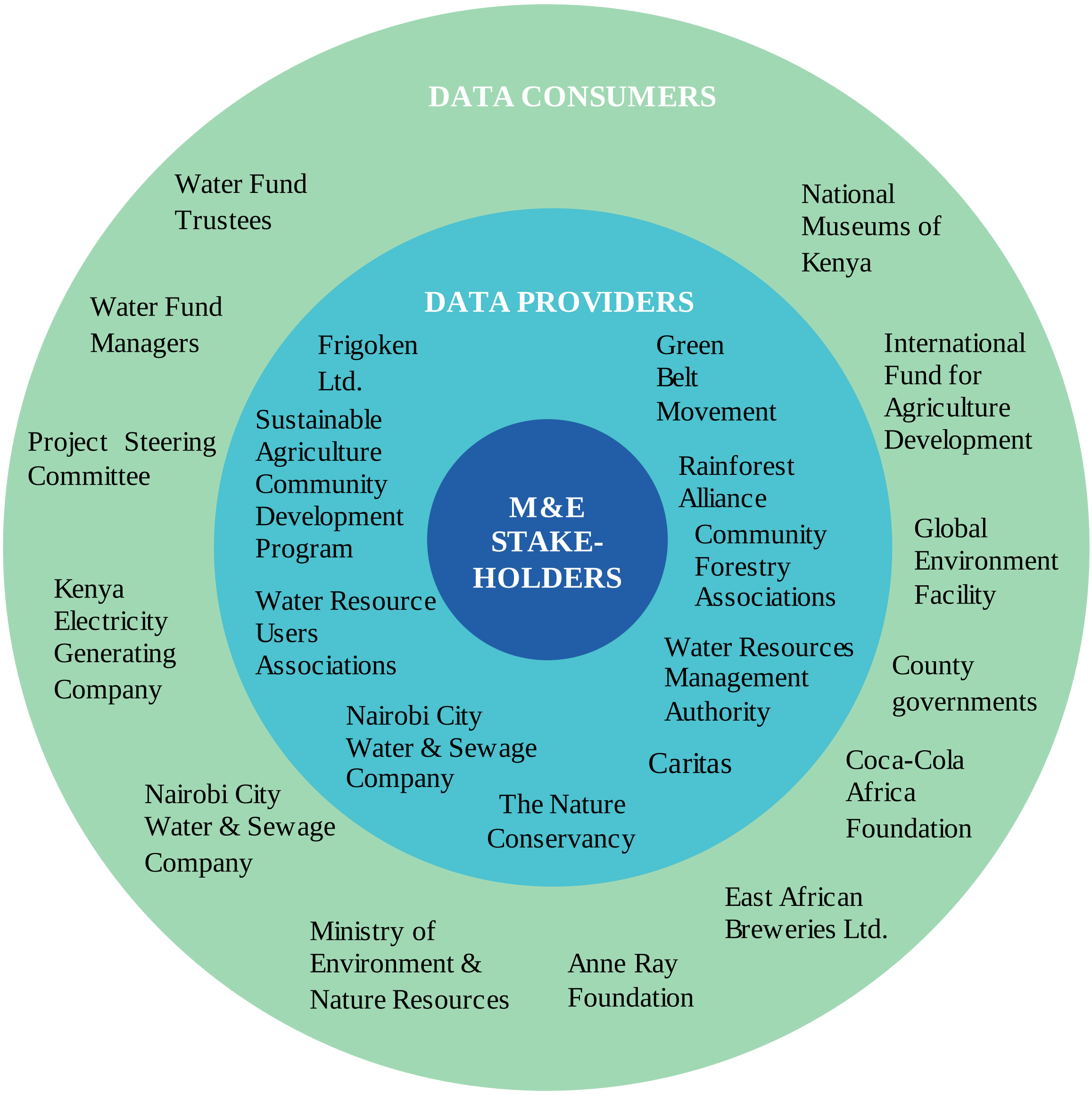


Figure 3: Stakeholder Monitoring and Evaluation Roles

Project Exit Strategy

The long-term goal is a water fund authority that is self-sustaining. To do this, the water fund needs to be ecologically, socially, and financially sustainable.

As export-orientated tea and coffee in the project area evolve due to market forces, new ecological water issues may arise, and thus a robust water quality monitoring system is required for long-term sustainability. Socially, the water fund depends on people in the upstream areas continuing to practice soil and water conservation measures. Given the large youth population in the project area, continued education about the benefits of soil and water conservation will be needed to maintain project gains. A danger is that pressure to use all available land for farming could result in riparian buffer zones and grass strips being planted with crops by those unaware of the benefits of these areas. Sustaining smallholders' awareness of the benefits of soil and water conservation are the key to the social sustainability of watershed activities. Financially, the endowment fund for the water fund is expected to generate sufficient revenue to ensure the water fund has the needed funding to continue catalysing soil and water conservation activities in the upper Tana watershed and conducting M&E activities.

Over the five years of the project, the Project Management Unit will evolve into the water fund's management team. By the end of the project's five years, the water fund is expected to be financially sustainable and on the way to social and ecological sustainability.

The triggers for TNC's exit out of the day-to-day management of the water fund will be: (i) when the project ends; and (ii) the water fund's endowment has been fully capitalized for one year.

Risks

There are several risks to the project's M&E activities. Here we list these risks and follow good practice by looking at both the likelihood of the risk and the impact if the risk happens.

Risk	**Likelihood and Impact**	**Mitigation Measures**
Multiple major landslides occur in the upper Tana, the water intake for Nairobi is forced to close, and support for the water fund dries up due to its perceived failure	High likelihood High impact	Develop a probability model for landslides in the upper Tana based on geology, soil type, and slope and focus on stabilizing high probability areas
Insufficient M&E funding	Medium likelihood Medium impact	Secure M&E funding from other sources
Unreliable monitoring data	Low likelihood High impact	Annual data quality assessment and audits
Implementation partners do not provide timely data	High likelihood Low impact	Tie partner progress payments to delivery of M&E data
Change in water quality is below the minimum detectable level of the downstream monitoring stations	Medium likelihood Low impact	Add upstream monitoring stations as funding allows

M&E Timeline

Key monitoring and evaluation dates in the project:

Baseline MPAT survey completed	March 2017
M&E Plan finalized	March 2017
1st six-month Progress Report due	April 2017
LDSF surveys begun	May 2017
Baseline MPAT report finalized	June 2017
Water quality review / biomonitoring workshop	June 2017
LDSF report finalized	September 2017
1st RIMS report due	October 2107
M&E workplan for the next year due	October 2017
1st Annual Project Report due	October 2017
2nd RIMS report due	March 2018
2nd six-months Progress Report due	April 2018
M&E workplan for the next year due	October 2018
2nd Annual Project Report due	October 2018
3rd RIMS report due	March 2018
3rd six-months Progress Report due	April 2019

Mid-term review	April 2019
M&E workplan for the next year due	October 2019
3rd Annual Project Report due	October 2019
4th RIMS report due	March 2018
4nd six-months Progress Report due	April 2018
M&E workplan for the next year due	October 2020
4th Annual Project Report due	October 2020
5th RIMS report due	March 2021
5nd six-months Progress Report due	April 2018
Begin MPAT and LDSF surveys	May 2021
Socioeconomic, carbon and water assessments completed	November 2021
Terminal performance evaluation	December 2021

M&E Budget

M&E activity	Responsible parties	GEF funding (US$)	Cost sharing (US$)	Total budget (US$)	Timeframe
Inception workshop	Project Management Unit (PMU)	15,000	--	15,000	17 January 2017
MPAT baseline survey	PMU, contractors	70,000	10,000	80,000	Draft report done by 30 June 2017.
MPAT end-line survey	PMU, contractors	20,000	[40,000]	60,000	Start in May 2021, with final report completed by 30 November 2021. [Need to identify cost sharing contribution.] No mid-term as per Logframe in project doc.
WEAI baseline survey	TNC, IFPRI	--	--	--	Completed in June 2015
WEAI end-line survey	PMU, contractor	--	--	--	Will only be done if $100k in funding is found
LDSF baseline surveys in five sites	PMU, ICRAF	75,000	--	75,000	Draft report done by 30 September 2017
LDSF end-line surveys in five sites	PMU, ICRAF	75,000	--	75,000	Start in May 2021, with final report completed by 30 November 2021
Carbon accounting baseline survey using FAO's Ex-Ante Carbon Balance Tool (EX-ACT)	Contractor	--	--	--	Completed during project design (Appendix 11 of project docs)
Carbon accounting end-line survey using FAO's Ex-Ante Carbon Balance Tool (EX-ACT)	PMU, contractor	35,000	--	35,000	Start in May 2021, with final report completed by 30 November 2021. (Project appendices para. 26 calls for 'an initial and a final carbon stock assessment'. No mid-term.)
Support capacity needs assessments for national	PMU, contractors	40,000	--	40,000	No description of this activity in the project

M&E activity	Responsible parties	GEF funding (US$)	Cost sharing (US$)	Total budget (US$)	Timeframe
tracking of ecosystems, food security and livelihoods					appendices. Appears only in the M&E Budget.
Biodiversity assessment and mapping (i.e., 'mapping of freshwater wetlands, the production of a wetlands biodiversity atlas, and an assessment of freshwater biological resources with an emphasis on those with food and feed potential' para. 27)	PMU, National Museums of Kenya	150,000	--	150,000	During project life
Water-quality database	PMU, WRMA	20,000	--	20,000	During project life
Travel to relevant regional and international meetings for sharing lessons learned	PMU, GEF focal point	24,000	--	24,000	During project life
Steering committee meetings	PSC, PMU	30,000	--	30,000	Bi-annually
Field monitoring visits	PSC, PMU	15,000	--	15,000	During project life
Mid-term review	Independent evaluator, IFAD, PMU	25,000	--	25,000	April 2019
Terminal evaluation	Independent evaluator, IFAD, PMU	30,000	--	30,000	At least one month before the end of the project. Report to be submitted no more than 12 months after project completion
External audits	Independent auditor, IFAD	25,000	--	25,000	Annually
Knowledge management platforms (DHIS2, Ministry of Environment and Natural Resources, and National Museums of Kenya)	PMU, NMK, contractors	110,000	--	110,000	During project life
Feasibility studies in at least 2 other Kenyan water towers	PMU, contractors	80,000	--	80,000	Toward project end
M&E publications and dissemination	PMU	60,000	--	60,000	During project life
Equipment and material (for monitoring)	PMU	78,000	--	78,000	Within nine months following project start-up
Contingency costs (10%)		--	--	--	TO BE ADDED AFTER M&E BUDGET REVISION

M&E activity	Responsible parties	GEF funding (US$)	Cost sharing (US$)	Total budget (US$)	Timeframe
Total		977,000	110,000	1,087,000	

Other costs not included in the budget above
Measuring 200 farmers' water use one year before and one year after the installation of a drip irrigation systems and rainwater pans, and measuring water use of 200 control farmers for two years and then giving them a drip irrigation systems and rainwater pans and continuing to measure water use for another year.

Masinga reservoir sedimentation measurements at outflow of two sub-watersheds where the project is working = $50,000

Africa Water Funds Summit 2018. Prepare and present water fund M&E framework to Africa wide audience = $60,000

Other Sources of M&E funding not included in the budget above
Coke: $50,000 for M&E with $40,000 over 2016 and 2017, and $10,000 in 2018.
ARF: $20,000 to supplement biodiversity surveys and database establishment and $30,000 for Africa Water Funds Summit science session.

Appendix 1: Indicator definitions and data collection sources

Indicator	Unit of measure	Definition	Data collection source/method	Frequency	Responsibility	Info use & audience
Outputs Strategy 1 1. Status of endowment fund		Identifies the progress of the endowment from conceptualization to operation. The steps in the process are: Decision by the water fund Board of Trustees to establish an endowment; investment manual for the endowment fund developed and approved; bank account opened; custodial services provider appointed; operational manual completed; capitalization begins; water fund established as the legal owner of the endowment fund; investment managers appointed; auditors appointed; Board of Trustees oversight committee appointed; and endowment in operation.	Official documents including the minutes of Board meetings.	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	PMU	TNC, project donors, and government officials
2. Government policies and strategies that refer to the water fund as an incentive model	Number	Government documents that mention the water fund as an incentive model.	PMU staff will track documents that mention water fund and retain an electronic copy wherever possible for verification purposes.	Annually	PMU	TNC and project donors
3. Amount of funding in the endowment fund	USD	The amount of funds in the endowment fund bank account.	Bank statements	Annually	PMU	TNC and project donors

4. Meetings held by the Project Steering Committee	Number	A count of Project Steering Committee meetings.	Minutes of Project Steering Committee meetings	Annually	PMU	TNC and project donors
5. Meetings held by the Board of Trustees	Number	A count of Board of Trustee meetings.	Minutes of Board of Trustee meetings	Annually	PMU	TNC and project donors
6. National and county advisory committee meetings	Number	A count of county advisory committee meetings.	Minutes of county advisory committee meetings	Annually	PMU	TNC and project donors
Strategy 2 7. People/ individuals/ household members receiving project services (RIMS 1.8.2)	Number	Total number of people (males/females) who have received services or benefitted from project activities	Water fund implementation partners will keep records on: Data recorder name First name of beneficiary in household Gender Age Mobile phone number GPS location of the activity The number of people will be calculated by multiplying the number of households receiving project services by average household size in the project area.	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	TNC, project donors, and government officials and media

			Data will be disaggregated by beneficiary gender and age			
8. Extent of land with rehabilitated or restored ecosystem services (RIMS 1.1.17)	Ha	This indicator measures progress made in rehabilitating, restoring, sustaining and enhancing the productive and protective functions of the land and natural ecosystems in the project area. Ecological restoration is defined as 'an intentional activity that initiates or accelerates the recovery of an ecosystem with respect to its health, integrity and sustainability. Rehabilitation emphasizes the reparation of ecosystem processes, productivity and services, but does not entail the re-establishment of the pre-existing conditions'. Examples include: targeted farm and landscape management practices (e.g., reforestation, afforestation, improved rangeland management, erosion control, agroforestry, removal of non-native species and weeds, reintroduction of native species, etc.); and the establishment and management of ecological buffer zones to	Water fund implementation partners will keep records on: # of acres with new soil conservation measures # of acres with new water conservation measures # of acres converted to perennial crops # of acres converted to agro-forestry Acres will be converted to hectares for the RIMS report.	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	TNC, project donors, and government officials and media

		reduce the impact of climate hazards (e.g., flood retention zones, storm breaks, and groundwater recharge zones).				
9. People adopting technologies that reduce or sequester greenhouse gas emissions (RIMS 1.1.18)	Number (gender disaggregated)	This indicator refers to the number of individuals who adopt practices resulting in carbon sequestration through the enhancement and protection of carbon stocks in the biomass, both above ground (e.g., conservation/restoration of degraded ecosystems) and below ground (in the soil organic matter). It also counts those who change their land-use practices in the forestry and agricultural sectors (e.g., agro-forestry and conversion to perennial crops).	Water fund implementation partners will keep records on: # of households that add permanent grass buffer strips # of households that convert some or all of their land to a perennial crop # of household with new on-farm trees planted # of households with bio-gas units The focus is on individuals who were directly and indirectly affected by the project. If a project introduces a technology that serves a group or community, each individual belonging to that group or community should be counted. If an individual benefits from the introduction of different technologies (e.g., biogas for cooking and lighting plus training	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	TNC, project donors, and government officials and media

			for improved fertilizer use to reduce N emissions), he or she should be counted only once. Data will be disaggregated by beneficiary gender and age			
10. Individuals involved in climate risk management, natural resources management (NRM) or disaster risk reduction (DRR) activities (RIMS 1.6.10)	Number (gender disaggregated)	The indicator quantifies the people enabled to engage and/or participate in climate risk management activities, disaster risk reduction efforts and/or a collective shift towards less climate-sensitive livelihoods. This indicator counts the number of beneficiaries who adopt more resilient technologies, such as diversification of farming systems (e.g., introduction of high-value, off-season crops) or the expansion of livelihood options	Water fund implementation partners will keep records on: # of households who implement soil conservation measures # of households who implement water conservation measures # of households that convert some or all of their land to a perennial crop # of household with new on-farm trees planted The number of households will be multiplied by the average household size in the project area (4.9) to estimate the number of individuals involved.	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	Reporting to IFAD

			Data will be disaggregated by beneficiary gender and age			
11. Smallholder household members supported in coping with the effects of climate change (RIMS 1.8.6)	Number	All household members that directly or indirectly benefit from climate change adaptation measures to address climate-related problems and risks, and that have earmarked budget allocations to support such adaptation measures. Such measures can include the improved analysis of climate-related risks and vulnerabilities; innovative technologies to respond to new and emerging risks; or the explicit scaling up of sustainable agriculture, land and water management practices (such as agroforestry, conservation agriculture, sustainable rangeland management, watershed management, erosion control, water harvesting or efficient irrigation systems).	Water fund implementation partners will keep records on: Data recorder name First name of beneficiary in household Gender Age Mobile phone number GPS location of the activity The number of smallholder household members will be calculated by multiplying the number of households receiving project services by average household size in the project area. Data will be disaggregated by beneficiary gender and age	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	TNC, project donors, and government officials and media
12. Number of acres of land reforested		This indicator counts the total number of acres of public land that has been replanted or restored after it had been reduced by fire or cutting.	Water Fund implementing partners will keep records on:	Quarterly	Implementing partners	TNC, project donors, and government officials and media

			• Area (acres) of forests rehabilitated • Number and species of seedlings planted Survival rate			
13. Households with installed rain water pans	Number	This refers to the households in the project area who have constructed water pans as a means of surface run off and/or roof water harvesting for irrigation purposes to improve food security and incomes	Water Fund implementing partners will keep records on: • Number of households with water pans • Volume of the water pans • Acres of land being irrigated/farmed	Quarterly and annual reports	Implementing partners	TNC, project donors, and government officials and media
14. Households with installed drip irrigation system	Number	This indicator will measure the number of households who have installed and are using drip irrigation systems in their farms to improve food security and incomes. Drip irrigation involves controlled delivery of water directly to individual plants through a network of tubes or pipes at very low rates (2-20 litres/hour).	Water Fund implementing partners will keep records on: • Number of households with installed drip kits • Acres of land being irrigated/farmed • Species of crops being farmed	Quarterly and annual reports	Implementing partners	TNC, project donors, and government officials and media

Strategy 3 15. Land Degradation Surveillance Framework (LDSF) surveys completed	Number	The LDSF is designed to provide a biophysical baseline at landscape level, and a monitoring and evaluation framework for assessing processes of land degradation and the effectiveness of rehabilitation measures (recovery) over time. This indicator will track the number of LDSF surveys that have been completed	Completed surveys by the TNC implementing partners	Annually	TNC and ICRAF	TNC, project donors, and government officials and media
16. Water monitoring stations installed or upgraded	Number	Water fund is conjunction with WRMA are using hydrometric gauging stations to provide information on water quality and quantity. This indicator will track the number of the gauging stations that have been installed and/or upgraded.	TNC and WRMA will collect and keep this data and will include: • Number and location of the stations upgraded • Number and location of stations upgraded	Annually	TNC and WRMA	TNC, project donors, and government officials and media
17. County Integrated Development Plans that reference the Multidimensional Poverty Assessment Tool survey results	Number	This indicator will track the number of counties who will refer the results of MPAT baseline survey when developing their policy documents including County Integrated Development Plan (CIDPs).	TNC will collect the data on: Number of policy documents per county referring to MPAT survey results.	Annually		TNC, project donors, and government officials

18. Information sharing platforms established	Number	The project will establish various information centers for storing, processing, and retrieving information for dissemination at regular intervals to project stakeholders. The indicator will measure the number of centres that have been established.	TNC and its implementing partners will keep information on: • Number of centres established • Number of visitors accessing the information	Quarterly and annual reports	TNC and its Implementing partners	TNC, project donors, and government officials and media
19. Meetings, conferences, and seminars held at national, regional and international levels where water fund knowledge and learning were shared	Number	The indicator will track the number of forums held (meetings, seminars and conferences) to share knowledge gained from implementing water fund.	PMU will keep the record of the forums held and type of knowledge shared.	Quarterly and annual reports	PMU	TNC, project donors, and government officials and media
20. Schools participating in school awareness programme	Number	This indicator measures the number of schools in the project area which are participating in the school awareness programme. The school awareness programme is an initiative of the project where schools which participate in the Ndakaini marathon are encouraged to form environmental clubs and given seedlings to plant in their schools and take care of them.	Water Fund implementing partners will keep records on: • Number of schools participating • Number and species of seedlings distributed	Quarterly and annual reports	Implementing partners	TNC, project donors, and government officials and media

21. New policies that give soil and water protection guidance on rural roads construction, quarries, afforestation, and riverine protection	Number	The indicator will measure the number of policies by county and national governments giving guidance on soil and water conservation, rural road construction, quarry management, afforestation and riverine protection.	• The policy documents • TNC will collect the data on the number of policy documents per county that give soil and water protection guidance.	Annually		TNC, project donors, and government officials
Outcomes 1. Effectiveness: increased ability of people to manage environmental and climate-related risks (RIMS 2.6.5)	Number	This is the assessment of the extent to which the project succeeded in building human capacity to manage short- and long-term climate risks and reduce losses from weather-related disasters. This will count the number of individuals engaged in climate risk management, environment & natural resource management (ENRM) and disaster and risk reduction (DRR) activities as per RIMS 1.6.10. The effectiveness rating will be based on the comparison of the actual achievement with the targets. A high percentage of achievement against the target will be considered as highly or satisfactory effectiveness of the	Water fund implementation partners will keep records on: # of households who implement soil conservation measures # of households who implement water conservation measures # of households that convert some or all of their land to a perennial crop # of household with new on-farm trees planted The number of households will be multiplied by the average household size in the	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	Implementation partners	Reporting to IFAD

		project initiative towards a human capacity goal.	project area (4.9) to estimate the number of people with increased ability to manage environmental and climate-related risks. Data will be disaggregated by beneficiary gender and age			
2. Tonnes of greenhouse gas emissions (CO_2) avoided and/or sequestered (RIMS 2.1.9)	Tons of carbon dioxide equivalent(CO_2e)	This is the assessment of the extent to which the project succeeded in avoiding or reducing GHG emissions (CO_2e).	Data will come from remote sensing carbon estimates using EX-ACT and the Land Degradation Surveillance Framework surveys	Baseline and end line	PMU, ICRAF	TNC, project donors, and government officials and media
3. Project lessons reflected in government policies, strategies or programmes (added by IFAD)	Number	Government documents that reflect the lesson learned by the water fund. This will include learning about soil and water conservation issues in the upper Tana River and successful incentives for soil and water conservation.	Government documents that the project staff see or hear about. Electronic copies of the documents should be retained for verification purposes wherever possible	Annually	PMU	TNC, project donors, and government officials and media
4.Days per year with turbidity measuring less than 5, more than 200, and more than 2,000 NTU at Nairobi's water intake	Number	Measurement of turbidity levels at the Nairobi water intake. Count of days for each of the three levels. Average daily NTU levels for each quarter sorted by levels of <5, >200 and >2,000 NTUs.	Nairobi City Water and Sewage Company data. They will share data with the project once a quarter.	Quarterly data submission and reporting to IFAD in 6-month Progress Reports	PMU and NCWSC	TNC, project donors, and government officials and media

5. Turbidity (NTU) in dry and wet seasons for both sub-watersheds and micro-watersheds	NTU	Monthly average turbidity levels in each of the 26-monitored sub and micro-watersheds.	Project-supported water quality monitoring station data loggers. PMU staff analyse the data periodically.	Bi-annually	PMU	TNC, project donors, and government officials and media
6. Sediment load in dry and wet seasons flowing into Masinga reservoir	Grams/litre	Total Suspended Solids (TSS) on the Maragua and Sagana Rivers just upstream of the Masinga reservoir	Water monitoring stations to be installed and TSS data to be collected periodically in conjunction with KenGen.	Bi-annually	PMU	TNC, project donors, and government officials and media
7. Change in water abstracted from a river by smallholder farming households before and after installing drip irrigation and/or a rainwater pan	%	The difference between the average amount of water used annually by treatment and control groups. A stepped-wedge design will be used whereby half the control households become treatment households after the first year and then the other half after the second year. The numerator = the average water usage per household in the control group. The denominator = the average water usage per household in the treatment group.	Water use logs kept by households and collected every quarter by PMU staff.	Quarterly for three years	PMU	TNC, project donors, and government officials and media
8. change in crop productivity before and after installing drip	%	The difference between the average amount of crop produce per acre annually by treatment and control groups. A stepped-wedge design will be	Crop productivity logs kept by households and collected after the harvest season each year by PMU staff.	Annually for three years	PMU	TNC, project donors, and government officials and media

irrigation and/or a rainwater pan		used whereby half the control households become treatment households after the first year and then the other half after the second year. The numerator = the average crop productivity per acre in the control group. The denominator = the average crop productivity per acre in the treatment group.				
9. Households with improved MPAT score	%	The numerator = the number of households in the MPAT survey. The denominator = the number of households with higher MPAT scores since the baseline.	MPAT survey data entered into the MPAT Excel tool.	Baseline and end line	PMU	TNC, project donors, and government officials and media
10.Households saying permanent vegetation cover on their farm has increased	%	The numerator = the number of households in the MPAT survey. The denominator = the number of households saying the percentage of permanent vegetation cover on their land has increased since the baseline.	MPAT survey data with analysis by PMU staff.	Baseline and end line	PMU	TNC, project donors, and government officials and media
11.Households saying soil erosion occurs on their land		The numerator = the number of households in the MPAT survey. The denominator = the number of households saying soil erosion occurs on their land.	MPAT survey data with analysis by PMU staff.	Baseline and end line	PMU	TNC, project donors, and government officials and media

12. Change in average number of trees per acre on survey households' land	%	Number of trees on a household's land divided by the land size in acres.	MPAT survey data with analysis by PMU staff.	Baseline and end line	PMU	TNC, project donors, and government officials and media
13. Kilometers of rivers protected	Km	Metres along the riverbank with permanent vegetation cover since 2013 that is at least 2 meters wide. Both sides of the river need to be covered to count as protected.	Records kept by project implementation partners.	Annually	Implementation partners and PMU	TNC, project donors, and government officials and media

Appendix 2: Targets versus actuals

Hierarchy	YR1 Target	YR1 Actual	YR2 Target	YR2 Actual	YR3 Target	YR3 Actual	YR4 Target	YR4 Actual	YR5 Target	YR5 Actual
Goal: The Upper Tana-Nairobi Water Fund as a Public-Private Partnership increases investments flows for sustainable land management and integrated natural resource management in the upper Tana catchment										
21,000 small holder farmer households with improved food-security, climate change adaptation and resilience capabilities	3,150 farmers		7,000 farmers		10,500 farmers		14,700 farmers		21,000 farmers	
Number of tons of greenhouse gas emissions (CO_2) avoided and/or sequestered	2% reduction compared to baseline		4% reduction compared to baseline		6 % reduction compared to baseline		8% reduction compared to baseline		10% reduction compared to baseline	
Development objective: a well-conserved upper Tana River basin with improved water quality and quantity for downstream users (public and private; maintaining regular flows of water throughout the year; enhancing ecosystem										
21,000 smallholder farmer households adopt climate-smart SLM practices (gender and age disaggregated)	3,150 farmers		7,000 farmers		10,500 farmers		14,700 farmers		21,000 farmers	
Improved base flow in rivers (M^3/sec)	2% improvement over baseline		4% improvement over baseline		6 % improvement over baseline		8% improvement over baseline		10% improvement over baseline	
Reduced sediment load in river basins in wet season (kg/l/sec)	2% reduction compared to baseline		4% reduction compared to baseline		6 % reduction compared to baseline		8% reduction compared to baseline		10% reduction compared to baseline	
Outcome 1.1: Multi-stakeholder and multi-scale platform supports policy development, institutional reform, and upscaling of integrated natural resource management (INRM) and sustainable land management (SLM)										
Water fund endowment is operational	Water fund dully registered		Water fund by-laws legally		Water fund capitalized by 50% of		Water fund capitalized by 75% of		Water fund capitalized by 100% of	

Hierarchy	YR1 Target	YR1 Actual	YR2 Target	YR2 Actual	YR3 Target	YR3 Actual	YR4 Target	YR4 Actual	YR5 Target	YR5 Actual
			established		target funds		target funds		target funds	
Relevant government policies and strategies refer to the water fund as an incentive model	-		At least 1 policy and strategies at national/ county levels		At least 2 policies and strategies at national/county levels		At least 3 policies and strategies at national/ county levels		At least 4 policies and strategies at national/ county levels	
Outcome 1.2: Policies and incentives support climate smart smallholder agriculture and food value chains in financially viable and sustainable watershed stewardships										
Coordinated watershed management policies at county and national levels	-		Input into 3 County Integrated Development Plans and sectoral strategies		3 County Integrated Development Plans coordinated with the WRMA		-		-	
Smallholder people/ individuals/ household members receiving project services/incentives (RIMS 1.8.2)	3,150 farmers		7,000 farmers		10,500 farmers		14,700 farmers		21,000 farmers	
Outcome 2.1: Increased land area, freshwater, and agro-ecosystems under INRM and SLM										
Number of hectares of land brought under climate-resilient management	15,000 Ha		30,000 Ha		50,000 Ha		75,000 Ha		100,000 Ha	
Number of individuals provided with technologies that reduce or sequester greenhouse gas emissions	3,150 farmers		7,000 farmers		10,500 farmers		14,700 farmers		21,000 farmers	

Hierarchy	YR1 Target	YR1 Actual	YR2 Target	YR2 Actual	YR3 Target	YR3 Actual	YR4 Target	YR4 Actual	YR5 Target	YR5 Actual
Outcome 3.1: Institutions capacitated to monitor Global Environmental Benefits										
Global Environmental Benefits monitoring tools and protocols integrated with partner institutions	1 Land Degradation Surveillance Framework (LDSF) surveys updated/ completed ≥ 10 water monitoring stations upgraded/ operational. WRMA prepared to house water-quality database		2 LDSF surveys updated/ completed ≥ 15 water monitoring stations upgraded/ operational.		3 LDSF surveys updated/ completed. 26 water monitoring stations upgraded/ operational. A water-quality database established and integrated into WRMA system.		4 LDSF surveys for at least 5 sub-watersheds updated/ completed.		5 LDSF surveys for at least 5 sub-watersheds updated/ completed.	
	Outcome 3.2: Monitoring and assessment framework supports the integration of climate resilience into policy making									
Multidimensional Poverty Assessment Tool (MPAT) survey results referenced in county development plans	MPAT survey result referenced by 1		MPAT survey result referenced by 3		MPAT survey result referenced by ≥ 3		MPAT survey results referenced by ≥ 4		-	

Hierarchy	YR1 Target	YR1 Actual	YR2 Target	YR2 Actual	YR3 Target	YR3 Actual	YR4 Target	YR4 Actual	YR5 Target	YR5 Actual
	County Integrated Development Plan		County Integrated Development Plans		County Integrated Development Plans		County Integrated Development Plans			
	Outcome 3.3: Knowledge management and sharing of lessons learned is facilitated									
Information sharing platforms established	-		1 county-level info centre		1 national-level info centre		1 info centre at National Museums of Kenya		-	
Number of meetings, conferences, and seminars held at national, regional and international levels where water fund knowledge and learning were shared	Inputs/ presentations at 1 meeting at either national, regional and international levels		Inputs/ presentations at 3 meetings at national, regional and international levels		Inputs/ presentations at 5 meetings at national, regional and international levels		Inputs/ presentations at 7 meetings at national, regional and international levels		Inputs/ presentations at 10 meetings at national, regional and international levels	
Lessons learned out-scaled to at least 2 other catchment areas in Kenya	-		Compiled lessons learned for scaling out		At least 1 feasibility studies conducted		At least 2 feasibility studies conducted		At least 3 feasibility studies conducted	

Appendix 3: Water quality monitoring

This water quality monitoring plan was developed in 2014 and has been under implementation (with some modifications) since 2015.

Methods

Station Selection – Sub-watersheds and Micro-watersheds

Because of different hydrologic characteristics, intended use of the data, and especially the human resources needed to collect samples, sample design and methods are divided between the two types of watersheds: sub-watershed and micro-watershed.

Sub-watershed water bodies are relatively large and will require specialized equipment for automated continuous data collection. These larger water bodies also will require rather advanced techniques for stream gaging and monitoring and therefore engaging professional hydrologists from within the Kenyan Water Resources Management Authority (WRMA), the Nairobi City Water & Sewage Company (NCWSC), and The Nature Conservancy (TNC).

Micro-watershed waters are small (~ <2 km^2) and wadeable at all but the highest flows. Because sampling will be done manually, equipment needs for micro-watersheds are minimal and sample collection will be done by trained community members in response to storm events. Training and QA/QC oversight will be by professionals.

At a planning meeting of project partners, June, 2014, consensus was developed around a set of 26 potential monitoring stations that would contribute to meeting the project goals. Table 1 and Figure 1 indicate locations and qualities of the suite of 26 potential stations.

Table 1 – List of potential stations and their respective priority as identified by project partners, June 2014.

Stn Code	Stream Name	Main River	Stn Type	Priority	Latitude	Longitude
4CB05	Thika	Thika	Sub-watershed	Medium	-0.810268	36.817916
4AD06	Gikira	Gura	Sub-watershed	High	-0.572920	36.917560
4BD07	North Mathioya	Mathioya	Sub-watershed	Low	-0.628623	36.954873
4AA05	Sagana	Sagana	Sub-watershed	Low	-0.445868	37.043460
4AA02	Thego	Sagana	Sub-watershed	High	-0.345870	37.047700
4AA01	Sagana	Sagana	Sub-watershed	High	-0.361880	37.068150
4AD01	Gura	Gura	Sub-watershed	High	-0.524100	37.018910
4BE01	Maragua	Maragua	Sub-watershed	Medium	-0.749508	37.157435
4BD01	Mathioya	Mathioya	Sub-watershed	Low	-0.714290	37.180104
GITHI01	Githika	Thika	Sub-watershed	High	-0.798901	36.834402
GITHA01	Githanja	Maragua	Sub-watershed	High	-0.789840	37.129220
KIAMA01	Kiama	Thika	Sub-watershed	High	-0.831903	36.818146
KIMAK01	Kimakia	Thika	Sub-watershed	High	-0.854390	36.827930
CHANI01	Chania	Thika	Sub-watershed	High	-0.851560	36.800862
MUMWE01	Mumwe	Gura	Sub-watershed	Medium	-0.517287	36.940884
MBOGI01	Mbogiti	Kiama	Micro-watershed	High	-0.839770	36.828120
KARAN01	Karangi Kiana	Kiama	Micro-watershed	High	-0.825950	36.803020

THIKA01	Thika Valley	Thika	Micro-watershed	High	-0.791760	36.805480
GITHAM01	Githambara	Githanja	Micro-watershed	High	-0.809880	37.072020
KAHAR01	Kaharo	Githanja	Micro-watershed	High	-0.796240	37.086750
KAMAH01	Kamahuri	Thego	Micro-watershed	High	-0.297890	37.112500
IRURI01	Iruri	Sagana	Micro-watershed	High	-0.363540	37.087370
GATUB01	Gatubia	Mere	Micro-watershed	High	-0.277110	37.111610
MITAI01	Mitai	Chinga	Micro-watershed	High	-0.584070	36.906050
MAROG01	Marogoya	Thuti	Micro-watershed	High	-0.561690	36.875560
GATHA02	Gathanji	Mumwe	Micro-watershed	High	-0.528528	36.862184

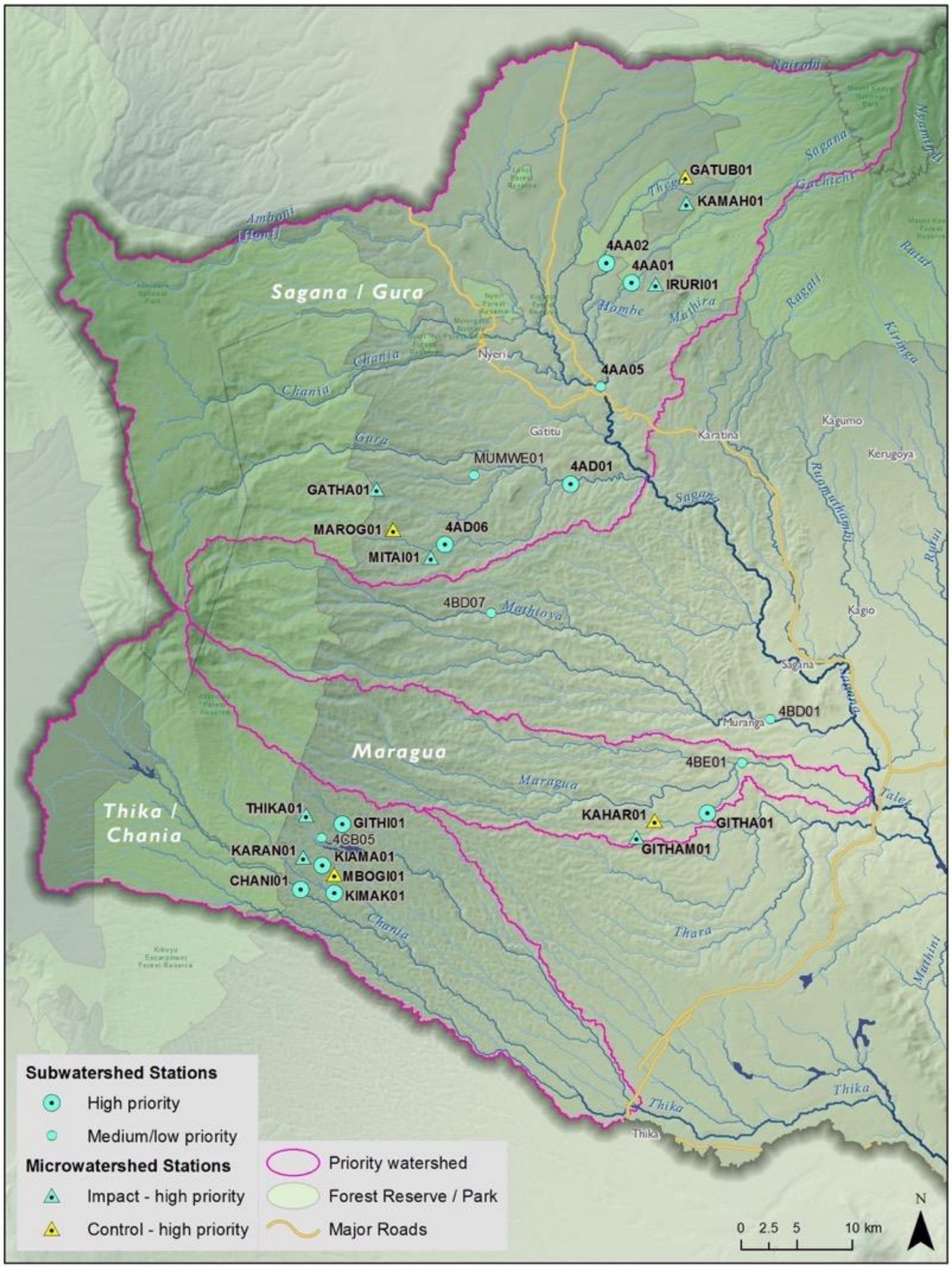


Figure 1 - Location of stations and type

Of the potential stations listed and shown above, 9 sub-watershed stations and 11 Micro-watershed stations were identified as high priorities and selected for monitoring during this phase of the project. Detailed descriptions of the high priority stations are provided in Appendices A and B.

Sub-watersheds

The goal for sub-watershed monitoring is to provide real-time information to water managers and to develop a long-term record of flow, suspended sediment, and storm event behaviour to identify trends in response to larger scale environmental conditions. Results are intended to help identify, refine, and prioritize management measures as well as guide the eventual development of E-flows. Table 2 lists the stations with their respective agency responsible for maintaining and sampling. NCWSC will also continue to maintain its daily recording at the Ngethu treatment works of system demand from the Chania-Thika watershed and water quality (turbidity).

Table 2 – Selected sub-watershed, monitoring station names and responsible parties

River Name	Station Name	Responsible Entity
Chania	Chani01	NCWSC
Kimakia	Kimak01	NCWSC
Kiama	Kiama01	NCWSC
Gikira	4AD06	WRMA
Githanja	Githa01	WRMA
Gura	4AD01	WRMA
Sagana	4AA01	WRMA
Thego	4AA02	WRMA
Githika	Githi01	WRMA

Equipment

At a minimum, each sub-watershed station will have a staff gauge located above a suitable permanent water level control point. Automated samplers such as Greenspan MP47, or equivalent, will be deployed in the main current to record stage and turbidity at pre-programmed intervals. Sondes may be placed in stilling wells as long as the sensors are exposed to ambient river conditions. In some locations, existing equipment will continue in use while at others newer equipment will be installed to upgrade or establish a station. Rating curves will be continued or developed following USGS (Turnipseed and Sauer, 2010) or equivalent methods.

Where the site and gaging station equipment are secure and there is a management benefit to real-time data reporting, some stations may be equipped with telemetry for both stage and turbidity. Given the cost of the equipment and vulnerability to vandalism and theft, only those stations where there is reasonable assurance of protection will be telemetered.

Instrument Sample Frequency

Sample frequencies will vary according to management needs, equipment constraints (e.g. battery capacity), water quality, antecedent conditions, and special events. Initially, we recommend sondes be programmed to collect data at 30 minute intervals. There will also be times when individual storm events or portions of a storm (e.g. ascending leg of the hydrograph) may require greater resolution and intervals will be shorter. Between the rainy seasons and/or during baseflow, sample intervals may be longer. Actual sample intensity will be determined on a case-by-case basis as guided by data and experience.

Manual Grab Samples

To supplement, validate and provide QA checks on instrumented data, staff gauge stage will be recorded and grab samples will be periodically collected for both turbidity and Total Suspended Sediment (TSS) at all stations on each visit. Grab samples will be collected as near to the centre of the channel as practical using a bucket on a line or by hand if safe. At least 2 litres of water will be placed in a clean container and labelled by station name, time and date. Field sheets (Appendix B) will be filled out that contain Station, Date, Time to nearest minute so results may be compared to the continuous record collected by the water quality sonde, Stage at time of collection, Sampler name, observations/comments, and space for any field results. We suggest initially collecting and measuring 3 or more samples manually within a sonde interval to determine instream variability within sample intervals at each station. This may need to be repeated at other flows.

Turbidity

Using a hand held turbidimeter calibrated to standards, turbidity may be analysed either immediately in the field or later in the lab. When pouring off a subsample aliquot for analysis, it is critical that the sample be well mixed to ensure that the results are representative of the sample.

Total Suspended Solids (TSS)

One-litre subsamples for TSS will be placed in labelled cubitainers and transported to the lab for analysis using Method 2540-D (APHA, 1995), USEPA Method 160.2, or equivalent. As with turbidity, it is extremely important that any subsample aliquot must be drawn from a well-mixed sample. Note that a special study is recommended (see below) that may enable reducing the number of TSS samples required. However, there will always be a need for some level of TSS sampling throughout the project for QA purposes. The frequency of TSS sampling will periodically be re-evaluated as results come forward.

Two Field Sheet formats for sub-watersheds are suggested in Table 3 below. The first form could be used if multiple observations at the same station are being collected on the same day. For example, if a single storm event is repeatedly being monitored throughout a day. The second format is better suited to data collected from multiple stations (e.g. a crew driving a circuit to service several stations in a region). The number of fields may be expanded as needed.

Table 3 – Field Form Options

Option 1

Station ID

Date	Time	Stage	TSS Bottle ID	Turbidity Sample ID or Result	Critical Observations	Initials of Sampler

Option 2

Station	Date	Time	Stage	TSS Bottle Number	Turbidity Sample ID or Result	Critical Observations	Initials of Sampler

Micro-watersheds

Micro-watersheds serve a different purpose from the sub-watersheds. These stations are intended to reflect water quality from very small (~ <2 km^2) watersheds. The theory is that the smaller the watershed, the more likely changes in land use practices will show short-term measureable change in water quality. This information can be used to demonstrate, assess, and direct conservation practices more immediately than may be possible at the sub-watershed scale.

Four clusters of micro-watersheds are arranged in sets within a larger sub-watershed; one or two treatments and one control, with similar soils, crops, and weather events. The treatment watershed represents an area in which conservation activities will be targeted toward sediment control. The control represents and area in which conservation activities are not planned. Table 4 and Appendix B show the Micro-watershed Clusters with their respective stations.

Table 4 – Micro-watershed Stations

Name	Cluster	Station Type	Latitude	Longitude	Elevation (M)
Mbogiti	1	Control	-0.83977	36.82812	2052
Karangi-Kiana	1	Treatment	-0.82595	36.80302	2156
Thika Valley	1	Treatment	-0.79176	36.80548	2199
Kaharo	2	Control	-0.79624	37.08675	1401
Githambara	2	Treatment	-0.80988	37.07202	1434
Marogoya	3	Control	-0.56169	36.87556	1980
Gathanji	3	Treatment	-0.52858	36.8622	2090
Mitai	3	Treatment	-0.58407	36.90605	1916
Gatubia	4	Control	-0.27711	37.11161	2086
Iruri	4	Treatment	-0.36354	37.08737	1848
Kamahuri	4	Treatment	-0.29789	37.1125	2073

Monitoring will be conducted for at least one year before conservation plans are implemented in order to develop the relationship between the two stations. This facilitates the requirements of a BACI design (see Resources section below for link to BACI theory tutorial). The goal will be to compare turbidity on representative storm events prior to and again after conservation interventions are installed.

Training

Micro-watersheds will be sampled manually by members of the local WRUA. We recommend that a single individual be identified as lead contact to be responsible for ensuring that monitoring is done. This could involve financial compensation or other incentive. Training will initially be required with regular follow-up sessions for QA/QC and support.

Sample Frequency

Sample frequency in the micro-watersheds will be determined by storm characteristics, especially intensity and antecedent flows. More frequent samples will be collected during the ascending leg of the hydrograph. At least 8-12 sample points should be collected per event, with the number modified as results indicate. In general terms, the goal is to gather enough data points to describe a variety individual storm types. Storm types should represent a range of runoff conditions. Examples might include, a runoff event from a single heavy rain of short duration, a prolonged rain and runoff event, a runoff event that follows a dry spell, an extreme runoff event. In all cases, an attempt should be made to sample during peak flow and peak turbidity. Note that these peaks may not be at the same time. In addition to storm events, periodic sampling at seasonal base flows should also be conducted twice monthly, at least initially to determine variability.

After conservation measures are implemented, runoff events representing similar conditions to the above storms will be targeted for sampling so that turbidity may be compared to "before" conditions. Figure 2 is an example of a before and after comparison without controls.

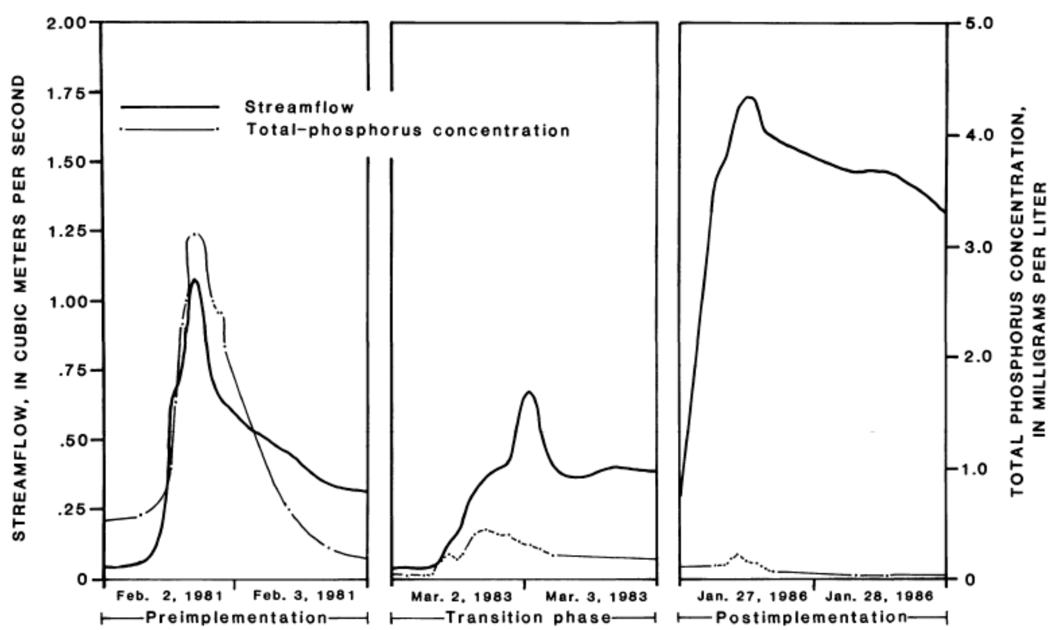


Figure 2 – Comparison of total phosphorus concentrations in three similar storms representing conditions before, during and after agricultural best management practices were implemented in a small watershed in Maine (Maloney and Sowles, 1987)

Stage
Stream stage reference points (e.g. staff gages, bolts, other immobile feature) should be sited upstream of each stream crossing. Points should be accessible during high flows however should be sufficiently above culverts and bridges to avoid influence from debris that artificially raises the stage near the culvert entrance. If a staff gauge is not used, reference points should be established that are higher than the highest anticipated water level so that measurements may be made during high water.

Sample Collection
Where it can be safely done, samples should be collected throughout the storm as flows are increasing and decreasing from near midstream just below the surface by either wading or using a pail suspended on a stick. Both time of collection and stream stage should be recorded. Care should be taken to avoid stirring up sediment above or in the area where the sample is to be taken. The sample should be taken by directing the sampler upstream into the current away from the person collecting the sample and approximately mid-depth if possible. Each sample should consist of at least 1 litre of water and uniquely labelled. Any floating debris (e.g. sticks, grasses, plastic, paper etc.) should be removed before saving the sample in its individual storage container. Intervals may be irregular but samples should be distributed to represent different flow and turbidity conditions such that both a hydrograph and turbidity graph may be constructed as in Figure 2 above.

Until or unless an acceptable relationship is established between turbidity and TSS, (see Special Study below) a 1-liter sample will be obtained, stored and sent to the lab for turbidity and TSS analysis. An example field sheet is provided in Table 5.

Table 5 – Example Field Sheet
Station Name____

Date	Time	Stage	Bottle ID	Observations	Initials of Sampler

Source Tracking within Micro-watersheds
Early on in the program, we recommend that a watershed survey be conducted during one or more runoff events. Micro-watersheds are small enough that the entire stream length may be walked in a few hours. The walk should begin at the micro-watershed sampling station (drain point) and proceed upstream noting obvious changes in turbidity, colour, flow, and location or presence of sources (e.g. runoff ditch, bank erosion, riparian condition) as noted either by GPS or sketched on a map. Photographs can be very helpful, GPS-linked if possible. Once the locations of water quality changes are documented, turbidity sources may be traced, either during the same storm or another or if the source is obvious (e.g., deep gully or ditch erosion). This information will be used to prioritize future conservation interventions as well as interpreting water quality change over time. We suggest that source tracking be done at least annually.

Laboratory Capacity - Total Suspended Solids (TSS)
TSS is time consuming, requiring multiple steps; drying filters (~12 - 24 hours), weighing filters to constant weight, filtering sample, redrying filters (another ~12 - 24 hours) and again weighing filters to constant weight. The process is therefore somewhat intermittent and not conducive to streamlined workflow.

We recommend that the WRMA Murang'a Laboratory, which is closer to the project area, be equipped and upgraded to analyse TSS using standard methods such as Method 2540-D (APHA, 1995), USEPA Method 160.2, or equivalent. Currently, WRMA Murang'a plans to expand its laboratory in the coming year. Staff at the WRMA Murang'a lab are fully capable but lack equipment and space to run TSS. Having the capacity to analyse for TSS in Murang'a would benefit the project. Table 6c is an estimate of costs to equip the lab for TSS analysis.

The importance of standardizing filter retention (pore) size across and throughout this study must be emphasized. Options vary, from filters that let relatively large particles pass to those that retain particles less than a micron. We recommend that 47mm Whatman GF/C glass fibre filters (or equivalent) be used. These retain particles down to ~1.2 uM. The filter apparatus budgeted below accepts the 47mm GF/C filter. If it is not practical to use these, at a minimum, whatever filter paper is eventually selected, it should be well documented and used for ALL analyses during this project to ensure comparability between and within sites.

Special Study - TSS-Turbidity Relationship
The water quality parameter of most interest to management for the Nairobi Water Fund is Total Suspended (and Settleable) Solids. The use of turbidity (a simpler, more rapid and less costly analysis) as a surrogate parameter for suspended solids (a more difficult and costly analysis) for water quality assessment is predicated on the assumption that a relationship between turbidity and suspended solids can be established. To facilitate sample analyses and especially to reduce laboratory expenses, we suggest that it is worthwhile to explore whether or not turbidity may be used as a surrogate for Total Suspended Sediment. To do so, a special study is needed to determine if a relationship exists between the two variables.

Because soil quality (esp. particle size, mineral composition and organic matter) affects light refraction (the actual measure of turbidity), separate relationships for each major soil type will likely be needed. As an initial trial, we suggest using one station from the Chania-Thika basin where red soils predominate and another in the Sagana basin where black cotton soils predominate. Stations may be those most convenient to access, but once selected, samples to develop the relationship must be taken from these alone.

Method
One litre of surface water from each station is collected from mid channel when water visually contains different levels of turbidity. Samples may be collected by wading or using a bucket on a rope from a bridge. Time of year and stage do not matter at this point. The goal is to collect at least 15 total samples from each station representing a range of conditions between clear water and extreme turbidity.

Samples are taken to and processed in the lab for both TSS and turbidity. Care is needed to ensure complete and equal mixing before aliquots for each sample are taken. Results are plotted to determine what, if any, correlation exists as the example shows in Figure 3. An R^2 >0.90 is a desired target.

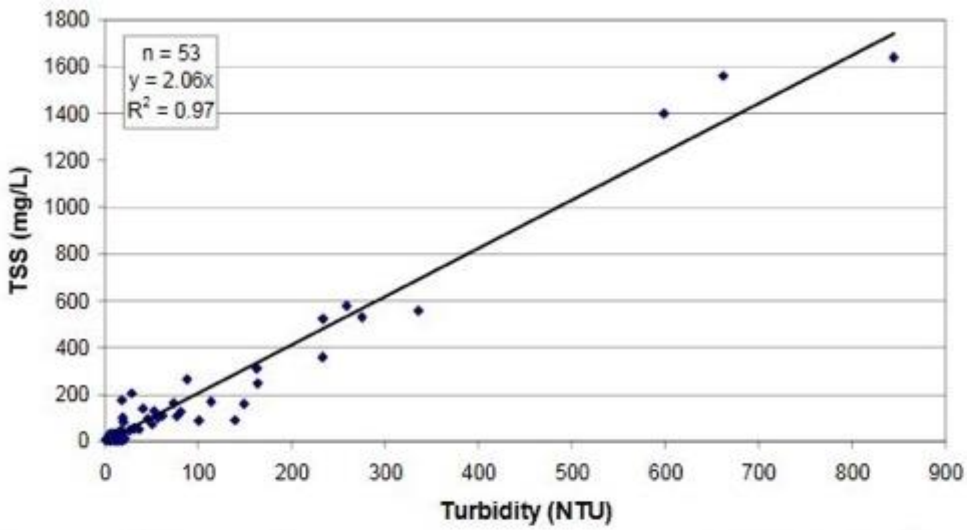


Figure 3 – Example of a relationship between sediment concentrations and turbidity at a USGS Station (n=53)

Depending on the composition of suspended sediment, especially if there exists a large portion of Settleable Solids, it may be advantageous to pre-treat samples using an Imhoff cone by allowing the heavy material to settle. A standard time (e.g. five minutes) is somewhat arbitrary, however, once decided, that settling time must be adhered to throughout the study. The supernatant is then used to develop the TSS-Turbidity curve, thus removing the particles that contribute to significant weight but do not contribute to light refraction. If this technique improves the correlation, then settleable solids will have to be measured separately. It may be possible that air drying the heavy material for 24 hours is sufficient.

Table 6 - Budget Estimate

Sub-watershed Field Equipment

#	Item	Cost Ea.	Total
9	Staff gauge	$100	$900
9	MP47 - turbidity and water level sonde	$5,820	$52,380
5	Telemetry module	$2,895	$14,477
18	1 qt Cubitainer (1-12pk)	$87	$1,566
1	Laptop	$800	$800
2	2100Q Portable Turbidimeters	$1,040	$2,080
Total			$72,203

Micro-watershed Field Equipment

#	Item	Cost Ea.	Total
11	Staff Gauge	$100	$1,100
5	Handheld Turbidimeter	$1,040	$5,200
24	1 qt Cubitainer (1-12pk)	$87	$2,088
11	Miscellaneous Supplies	$100	$1,100
Total			$9,488

Laboratory Equipment

Item	VWR #	Cost Ea	#	
Drying Oven	414005-114	$1,686.93	1	$1,686.93
PYREX® 2.4 L Small Knob Top Desiccator	89134-384	$229.48	1	$229.48
Plate	89038-068	$64.01	1	$64.01
Ohaus® Adventurer™ Pro Analytical Balance, 65 g	470020-258	$2,358.00	1	$2,358.00
Vacuum Pump and Compressor	470123-246	$524.09	1	$524.09
PYREX® 47 mm Microfiltration All-Glass Assembly, 300 mL Funnel - Complete.	71001-602	$438.09	3	1314.27
Suction Flask 1 L	89001-826	66.55	3	199.65
Manifold			0	
Forceps	89259-954	$6.48	3	19.44
Total-High				$6,395.87

Acknowledgements

Preparation of this document is really the result of a team effort on the part of all Upper Tana – Nairobi Water Fund Partners as well as all participants of the Monitoring Planning Seminar held June 9 and 10, 2014 in Murang'a. Special thanks is owed staff from Water Resources Management Authority and the Nairobi City Water and Sewer Company who have been extremely supportive and knowledgeable throughout development of the plan. Fred Kihara and Colin Apse, TNC Africa Program, Dave Courtemanch, TNC Maine, and Kirk Klausmeyer, TNC California all provided valuable comments and questions throughout making our task easy. Kirk's patience and diligence producing multiple iterations of maps and station codes was especially helpful.

Primary funding for this plan was generously provided by the Pentair Corporation.

Appendices

Appendix A – Sub-watershed Station Descriptions –in Dropbox as SubWatershedStnDescriptions.docx.

Appendix B – Micro-watershed Station Descriptions – in Dropbox as MicroWatershedStnDescriptions.docx.

References

American Public Health Association (APHA), American Water Works Association (AWWA), and Water Environment Federation (WEF) (1995). "2540 D: Total suspended solids dried at 103 – 105oC." In A. D. Eaton, L.S. Clesceri, and A. E. Greenberg (Eds.), *Standard Methods for the Examination of Water and Wastewater*, 19th Edition, Washington, D.C.

Maloney, T. and J. Sowles. 1987. Effects of agricultural best-management practices on total phosphorus yields in the Johnson Brook and Lovejoy Pond watershed, Kennebec County, Maine. -1980-1984. US Geological Survey Water Resources Investigations Report 87- 4118. Augusta, Maine.

TNC, 2014. The Upper Tana – Nairobi Water Fund Implementation Monitoring Plan 2013-2015 – June 2, 2014. The Nature Conservancy – Africa Program, Brunswick, Maine.

Turnipseed, D.P., and Sauer, V.B., 2010, Discharge measurements at gaging stations: U.S. Geological Survey Techniques and Methods book 3, chap. A8, 87 p. (Also available at *http://pubs.usgs.gov/tm/tm3-a8/*.)

Web Resources

Sampling, Gaging and Analysis

USGS Techniques of Water-Resource Investigations Reports <http://pubs.usgs.gov/twri/> (this is an older but comprehensive list of downloadable pdfs)

BACI design theory and tutorial

Many examples available on web through search on "BACI design." One easy tutorial is http://people.stat.sfu.ca/~cschwarz/Stat-650/Notes/MyPrograms/BACI/BACITalk-2012-10-15-UBC/baci.pdf

Hach turbidimeter operating manual and guidance

http://www.hach.com/2100q-portable-turbidimeter/product-downloads?id=7640450963

Total Suspended Solids

http://www.epa.gov/region9/qa/pdfs/160_2.pdf

Appendix 4: Outcome indicators selection

Choosing the outcome indicators[10] a project will measure is a critical choice. What one chooses to measure become how one defines success, and stakeholders need to agree with how success will be defined. All of the outcome indicators in this plan have been vetted by key project stakeholders to ensure they agree with how we are defining success.

For this plan, we prioritized three kinds of indicators critical to the sustainability of the water fund: (i) water quality and quantity indicators; (ii) watershed protection indicators; and (iii) human well-being indicators.

The water fund's *raison d'etre* is to improve water quality and quantity at the water intake for the city of Nairobi and two of the major rivers that feed the Masinga reservoir that powers a cascade of five hydroelectric dams on the upper Tana River. Water quality, and specifically turbidity in the wet season, matters greatly. When turbidity is high, the water treatment plans have to reduce the throughput to give the water more resident time for clarification. This reduces the amount of water flowing to the city. Decreasing the average level of turbidity also reduces the city's water treatment costs because cheaper alum flocculants can be used as long as the turbidity stays under 200 NTUs. Turbidity is also a critical concern for hydropower generation on the upper Tana because losses in reservoir capacity reduce long-term dry-season hydropower generation and reduce Kenya's climate change resilience.

Water quantity is also important because Kenya is a water-scare country, and Nairobi has a large population with limited or no access to municipal water supplies. When there is a drought like in 2009 and 2017, there is insufficient water in the rivers and reservoirs that supply Nairobi, and water rationing in the city is required. Increasing dry-season flows in the upper Tana benefits Nairobi during drought years and benefits the country's hydropower generation every year.

Costs were a key consideration for the water quality and quantity monitoring. Hence, we partnered with existing governmental and parastatal organizations already collecting water quality and quantity data in the project area and upgraded or installed 26 priority water monitoring stations in exchange for joint access to the water monitoring data the stations produce. The partner organizations have ownership and long-term operational responsibility for the stations. This partnership reduces the project's data collection costs and improves the long-term sustainability of the data monitoring.

Watershed protection indicators are vital for showing stakeholders how the project has improved soil and water conservation in the upper Tana. We chose our watershed protection indicators based on a baseline household survey in 2013 that highlighted indicators relevant to local soil and water conservation and known to vary substantially in the project area. These indicators include the percentage of sloping land with soil conservation or erosion control measures, percentage of land within 2 meters of a stream or river that has permanent vegetation, number of farmers who irrigate

[10] We ignore output indicators here because they track progress on agreed deliverables and are proscribed by the project activities. Selecting output indicators is relatively straight forward and getting one wrong is unlikely to impact the social, political or financial sustainability of the water fund.

using river water, overgrazing in the community, and how long does it take for the colour of the local river water to clear after a rain.

We chose human well-being outcome indicators that focus primarily on the benefits to households from increased crop yields and better food security. The GEF Implementing Agency for this project, IFAD, specified that we use their Multidimensional Poverty Assessment Tool. This tool covers: food and nutrition security, domestic water supply, health and heath care, sanitation and hygiene, housing, clothing, energy, education, farm assets, non-farm assets, exposure and resilience to shocks, and gender and social equality. In addition to crop yields and better food security, this tool will identify positive and negative outcomes in unanticipated areas. It is a peculiar tool however…